

From Silos to Systems: Process-Oriented Hazard Analysis for AI Systems

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Abstract

To effectively address potential harms from Artificial Intelligence (AI) systems, it is essential to identify and mitigate system-level hazards. Current analysis approaches focus on individual components of an AI system, like training data or models, in isolation, overlooking hazards from component interactions or how they are situated within a company’s development process. To this end, we draw from the established field of system safety, which considers safety as an emergent property of the entire system, not just its components. In this work, we translate System Theoretic Process Analysis (STPA) – a recognized system safety framework for analyzing AI development and operation processes. We focus on systems that rely on machine learning algorithms and conduct STPA on three case studies involving linear regression, reinforcement learning, and transformer-based generative models. Our analysis explored how STPA’s control and system-theoretic perspectives apply to AI systems and whether unique AI traits—such as model opacity, capability uncertainty, and output complexity—necessitate significant modifications to the framework. We find that the key concepts and steps of conducting an STPA apply to AI systems but require targeted adaptations to address AI-specific challenges—such as model opacity, autonomy, and output complexity—that arise to differing degrees across AI paradigms such as linear models, reinforcement learning, and generative models. We present the Process-oriented Hazard Analysis for AI Systems (PHASE) as a guideline that adapts STPA concepts for AI. Applying and interpreting STPA using the PHASE guidelines enables four key affordances for analysts responsible for managing AI system harms: 1) detection of hazards at the systems level, including those from accumulation of disparate issues; 2) explicit acknowledgment of social factors contributing to experiences of algorithmic harms; 3) creation of traceable accountability chains between harms and those who can mitigate the harm; and 4) ongoing monitoring and mitigation of new hazards.

CCS Concepts

• **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → **Empirical studies in HCI**.

Keywords

System safety, Sociotechnical AI Safety, Hazard Analysis, Responsible AI Development, System Theory, Control Theory, System-level Analysis, System Theoretic Process Analysis

1 Introduction

When it comes to safety and risk management of Artificial Intelligence (AI)-based technologies, there is an emerging recognition that hazards - potential sources of harm - from such technologies need to be viewed and analyzed at the (sociotechnical) system level [26, 80]. Regulatory efforts such as the European Union AI Act explicitly distinguish between the terms AI model and AI system [27]. In parallel, major AI developers are releasing system cards in addition to model and data cards, recognizing that the performance of many recently released Application Programming Interfaces (APIs) is the result of the amalgamation of multiple models and software components [4, 49]. Additionally, the role of the algorithmic supply chain, which includes the actors involved in developing, integrating, and using an AI system, is increasingly recognized as central to understanding how harms emerge from AI systems [5, 23]. Finally, there is growing recognition of the importance of defense-in-depth approaches to mitigating risks from AI systems, which advocate for multiple mitigation mechanisms applied across the AI development lifecycle [11]. Despite growing views of AI-based technologies as sociotechnical systems, there has been surprisingly minimal empirical effort to develop and examine frameworks that facilitate hazard analysis at the system level for AI systems. In this work, we draw on a well-established hazard analysis framework from the field of system safety and examine how it could be applied to analyze the processes involved in developing and integrating an AI system.

Hazard analysis has a long-standing history in the field of safety engineering, and a track record of being an essential component of developing and deploying complex safety-critical technologies such as nuclear power plants and passenger planes [20, 36, 40]. Yet, the AI community has been slow at embracing the fundamental know-how and practices well established in safety engineering [20, 36, 40]. We specifically draw on the field of system safety, one of the most recent and significant developments in safety engineering [39]. System safety emerged from the observation that safe systems cannot be achieved by merely analyzing technical components in isolation. Instead, system safety emphasizes the necessity of considering the broader social and technological context. System safety frameworks provide the means of analyzing systems as a whole rather than conducting multiple, separate analyses of their composing sub-systems (e.g., machine learning (ML) model, development team, user) [43, 51].

In this work we examine to what degree a well-established hazard analysis framework can be readily applied to assess AI systems and what it affords to those conducting the assessment. In particular, we focus on STPA, a framework well-known for its utility in analyzing complex technical systems [41]. STPA offers two powerful affordances essential for examining hazards at the system level:

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1) It provides the means to map out and analyze complex social and technical systems, leveraging concepts from system theory and control theory. 2) It centers on the potential loss of stakeholder values, which facilitates harm analysis beyond safety-critical harms (i.e., loss of life, damage to property, and environmental harm). However, many of the current guidelines for conducting STPA primarily focus on traditionally safety-critical systems such as nuclear power plants. There are currently only a handful of examples of its application to AI systems and algorithmic harms [33, 60]. When applying STPA to AI systems, we build on existing work while focusing on characteristics and challenges unique to conducting hazard analysis for AI, such as system opacity, capability uncertainty, and output complexity, as well as the diverse range of potential harms that could result from their adoption. Crucially, these characteristics are not uniform across AI systems: a linear regression model, a reinforcement learning agent, and a transformer-based generative model differ substantially in their opacity, autonomy, and output complexity, and therefore place different demands on each stage of the analysis. We treat this heterogeneity as central rather than incidental to our study, and our case studies are chosen to expose it. Specifically, we ask the following research questions:

- RQ1: How can the steps of the STPA framework be translated to analyze potential hazards from AI systems?
- RQ2: What does STPA afford an analyst in identifying, assessing and mitigating potential harms?

We conducted three case studies spanning three AI systems (linear regression, reinforcement learning, transformer-based text-to-image models) across two domains of application (creative practice and medicine). Following the existing handbook by Leveson and Thomas, we conducted STPA on the three case studies [41] and documented our interpretive and reflexive experience in the process. We find that STPA-based analysis of AI provides the means for detecting potential sources of harm along both the development and operation of AI systems. We present a guideline, Process-oriented Hazard Analysis for AI Systems (PHASE), documenting our translation of STPA for AI, considering the aforementioned unique characteristics. The guideline introduces key considerations that need to be accounted for when applying STPA for AI systems. This work contributes to the growing body of responsible and safe AI practices discourse by: 1) producing PHASE guidelines that adapt STPA to the specific challenges of AI systems and enable system-level hazard analysis for AI; 2) providing three diverse case studies to illustrate the outcome of the analyses, and 3) presenting four affordances as shown by empirical investigation of these case studies.

2 Background

In this section, we situate our work in today’s AI safety discourse, Responsible AI (RAI), and system safety literature. We then briefly describe the key tenets and the steps of the STPA framework in Section 2.3.

Note that in this work, we use the Organisation for Economic Co-operation and Development (OECD) definition of an AI system: “a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.” [54]. An AI system consists

of multiple components, including the input dataset, model, and resulting output.

2.1 Examining Potential Sources of Algorithmic Harm

Alongside the increased public awareness of how AI systems can introduce societal disruptions and harm to individuals, RAI scholars have been actively studying the various types of algorithmic harms and means of harm mitigation. Shelby et al. [66], for example, present at least 170 articles related to sociotechnical harms of algorithmic systems – defined as “adverse lived experiences of individuals and communities.” The rapid advancements and adoption of large-scale, multi-modal, and general-purpose ML models in the last five years have further intensified the discussions about AI-related harms [10, 15, 69]. The emerging epistemic community of AI safety is also drawing more attention to concerns about AI-driven existential and catastrophic risks to humanity [3, 18, 61]. To date, the types of harm that can be expected from current and near-future AI systems remain a subject of heated debate.

Alongside these discussions are multitudes of theoretical frameworks, tools, and processes proposed to analyze and mitigate potential sources of such harms. Qualitative and normative frameworks are commonly used to identify societal, ethical, and legal factors, helping analysts reason about how AI systems may produce unfair, exclusionary, or otherwise harmful outcomes in specific contexts of use [47, 58, 79]. Complementing these approaches, technical tools, often implemented as software libraries or code bases, are employed to examine potential sources of harm within datasets and models, such as data quality issues, bias, and performance disparities [7, 53]. Safety and security engineering tools – inspired by processes such as red teaming [30] or Failure Mode and Effects Analysis (FMEA) – aim to anticipate how AI models may behave under adversarial, unexpected, or edge-case conditions, thereby identifying model-level failure modes that could lead to downstream harms [31, 44, 56, 59]; Additionally, participatory and community-based research methods have been adopted to uncover harms that emerge through real-world interactions between AI systems, individuals, and communities, particularly those that may be overlooked by purely technical evaluations [6, 16, 45, 82].

Existing methods are designed to be effective in analyzing hazards either by focusing on a single component of an AI system (e.g., representation in a dataset) or examining interaction instances (e.g., how much a decision maker relies on an AI assistant) in silos without examining their connections [75, 84]. As a result, these approaches do not support the kind of integrated, end-to-end reasoning about hazards that is central to system safety analysis.

2.2 Safety Engineering and System Safety for AI

Safety engineering frameworks have been used to establish the safety of technical systems as early as the 1940s with the use of FMEA and Fault Tree Analysis (FTA) in military technology and the Apollo mission [19, 20, 71]. These early frameworks follow an analytically reductive approach where component failures and human errors can be analyzed as a chain of events that lead to an incident. With the increase in complexity of technological systems

and specifically the introduction of software, the analytically reductive approach was deemed ineffective in capturing new types of hazards. In the early 2000s, Nancy Leveson proposed an alternative accident model called Systems-Theoretic Accident Model and Processes (STAMP) [39]. Consistent with system theory [76], STAMP frames safety as an emergent property of a system, resulting from the interactions and control structures among people, technical artifacts/agents, and the environment [40]. This paradigm of safety motivated the safety engineering community to look beyond the analysis of units and components, and consider analyzing the interaction between software programs, physical parts, and people needed for establishing a safe system [36, 42, 43]. Subsequently, Leveson and Thomas developed System Theoretic Process Analysis (STPA) as a *hazard analysis* method based on STAMP [41], which was promptly adapted across a number of safety-critical application domains [8, 22, 55].

Identifying and mitigating harms from AI systems requires a systematic approach that explores and maps out the distribution of existing product development practices beyond an ML model and across individuals, automated systems, collectives, and institutional mechanisms [26, 33, 37, 38]. Narrowly examining and designing for safety at the ML model level is important but not sufficient for ensuring the safety of products that only partially rely on these models [17, 35, 47]. The system theoretic perspective recognizes that AI models do not exist in silos in practical product environments. They are often integrated in a complex network of API calls, other AI models, a range of datasets and model outputs, and various user interfaces [24, 51, 83]. In addition, an AI system design is influenced by the decisions of a wide range of actors, including the data curation team, model development team, responsible AI analysts, users, company executives, product managers, and regulators [28, 48, 67].

Traditionally, STPA and system safety have been applied to technologies where there is consensus on what is safe or unsafe (i.e., an airplane that could crash is viewed as unsafe). In contrast, as discussed above, harms arising from AI systems are diverse, contested, and often context-dependent, with no single agreed-upon notion of acceptable or unacceptable outcomes. Accordingly, we treat potential harms from AI as emergent properties of the system that arise through interactions among technical components, human actors, and institutional processes across development and operation. Furthermore, AI systems have unique differentiating characteristics compared to other complex cyber-physical systems: they can have varying levels of outcome complexity, and the system capabilities [85] can be probabilistic and distributed. Importantly, these properties are not shared uniformly by all AI: they vary markedly across AI paradigms—for instance, between an interpretable linear model, an adaptive reinforcement learning agent, and an opaque generative model—so that the same analytical step can demand different considerations depending on the type of AI under study. These properties complicate hazard identification and prediction, and must be explicitly considered when applying system-theoretic hazard analysis. These considerations motivate the need to adapt STPA for AI systems in a way that supports system-level reasoning about harms across the AI development and deployment processes.

2.3 System Theoretic Process Analysis (STPA)

Below, we provide a brief description of the key steps for conducting STPA outlined in the STPA handbook [41]. These steps are meant to be performed iteratively and cyclically across the lifecycle of technology development.

2.3.1 Identify purpose of the analysis. The STPA process starts by identifying losses, system boundaries, and hazards. By definition, a **loss** occurs when a stakeholder’s value is disregarded or violated. **Hazards** are a *system state or conditions that could be a potential source of harm*. Hazards are defined per **system boundary**, which outlines parts of the system where designers, developers, engineers, or automated agents could exercise some level of control. To provide a simplistic example, a construction company repairing a roof of a house may point to safety considerations (loss of human life) to be important for the project; identify the possibility of workers falling from the steep and bare edges of the rooftop as a fall hazard; and is only able to install safety equipment such as fences within the boundaries of the roof being renovated and not that of the neighbouring house (system boundary).

2.3.2 Create control diagrams. The next step involves creating control diagrams to visually capture the model of the system within its system boundary. It represents the elements and interactions between different elements within the system in the form of a feedback loop called *control feedback loop*. The two key elements of a control feedback loop are the **controller** – an algorithm, a person, or an organization that can make decisions or perform actions to control something – and the **controlled process** – the component, process, or another controller being controlled by the controller. A controller is represented with a *control algorithm* – a description of how the controller makes decisions – and *process model* – the controller’s internal/mental model the environment used to make decisions.

The relationship between controllers and controlled processes are represented with two types of interactions: **control action**, which is a constraint that the controller puts on the controlled process, and **feedback**, which is the response/feedback the controlled process provides to the controller. Typically, the various controller, controlled processes, and their interactions are arranged visually in a hierarchical manner, such that those with the highest level of control/authority appear on top.

2.3.3 Identify unsafe control actions. Once a control diagram is drawn, the next step is to identify actions that could be taken by the controller that lead to any of the hazards identified earlier. These are called **Unsafe Control Action (UCA)**. An UCA is fully described using five parts: the controller (*source*) that provides the problematic action (*control action*), the *type* of action/inaction that could lead to hazards (e.g., is it the execution, lack of execution, untimely execution of the control action that can lead to the hazard?), specific system condition or state context (*context*), and *link to hazard*.

2.3.4 Identify loss scenarios. Finally, the circumstances that could lead to a UCA – **loss scenarios** – are identified. To identify loss scenarios, the analysts need to consider all potential issues with the elements and interactions, such as missing feedback loops, incorrect mental or process models, and improper control algorithms. At the

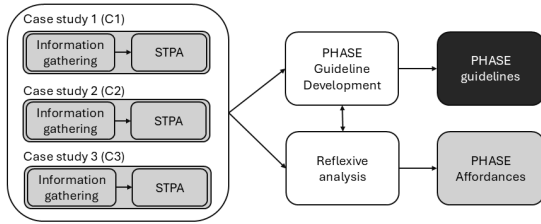


Figure 1: Methodology overview: STPA was conducted for each case study after we gathered the necessary information. The information gathering process was different for each case study. Iterative reflexive analysis led to the generation of the PHASE guideline and identification of affordances.

end of this process, the analysts develop a list of safety requirements based on the loss scenarios to provide the necessary constraints and to minimize or eliminate cases of inadequate control.

3 Methodology

Figure 1 presents an overview of our research method. As Njie and Asimirian outline, case studies as a research method enable in-depth and first-hand investigation of events and processes [52]. Therefore, we used case studies to investigate how STPA needs to be interpreted when it is used to analyze AI systems. Through an iterative, reflexive analysis of the case study applications, we developed PHASE, a guideline for conducting STPA on AI systems, and identified the key affordances it provides. We further refined the guidelines’ language based on feedback from a pilot study, described in the supplementary material. Here, we describe the case studies, the details of applying the STPA framework, and the process of developing reflexively analyzing PHASE.

3.1 Case studies

To illustrate the applicability of STPA we chose the following real-world case studies based on the diverse range of ML model they represent, application context, and the nature of interaction between the AI system and a human user. Specifically, the three cases span distinct ML paradigms—supervised linear modelling, reinforcement learning, and transformer-based generative modelling—and distinct interaction modes—human-in-the-loop, human-out-of-the-loop, and human-as-tool-user—so that we could examine whether and how different types of AI place different demands on the analysis rather than treating AI as a single, homogeneous technology. An additional selection criterion was our ability to work closely with the teams developing these systems, enabling access to design decisions and development processes that are essential for conducting such analysis.

- **Case 1 (C1) - Early warning system:** A linear regression algorithm is being designed in the Netherlands to predict the likelihood of late-onset sepsis (LOS) in pre-term infants before a clinical suspicion [74]. It is to be used as a proactive warning system alerting clinicians to start early diagnostic testing for LOS and, ultimately, provide timely treatment.

As a human-in-the-loop system, a clinician makes final, clinical decisions.

- **Case 2 (C2) - Insulin Injection:** An Reinforcement Learning (RL)-based, automated and personalized insulin injection system is being developed in Canada for patients with diabetes [9]. Some diabetic patients use an artificial pancreas that facilitates automatic insulin injection. The RL algorithm is being developed for use in the artificial pancreas to allow tailored insulin doses for each individual in a human-out-of-the-loop manner.
- **Case 3 (C3) - Storyboarding :** Some ML artists have adapted Text-to-image (T2I) models as part of their creative practice, such as storyboarding for video creation [29, 44, 63, 86]. In this case study, we focus on T2I demo platforms generally. While we do not focus on a specific platform, many such platforms are produced by companies in the United States, and rely on some combination of transformer-based or diffusion architecture T2I generative models.

3.2 Applying the STPA framework

To apply STPA, we started by gathering information about the AI system, its stakeholders, and how the system was developed and integrated in the given context. To do so, we connected with the relevant stakeholders of each of the projects and conducted secondary research. Considering time limitations, data availability, and access to subject-matter experts, the information-gathering process was customized for each case. For C1, the lead author held a preliminary meeting with the ML developer and data scientist to understand the project’s objectives and used a peer-reviewed academic publication on the project as the main reference for the analysis. For C2, the lead author held monthly meetings for eight months with two RL researchers – one of whom developed RL algorithms for insulin injection. For C3, the lead author conducted eight interviews and one workshop (15 participants) involving experts along the T2I model development process and ML artists who use these models in their creative practice.

After information gathering, the lead author followed the STPA steps outlined earlier (Section 2.3) and applied them for each case study. While the procedural steps are those of standard STPA, the interpretive work at each step—deciding what counts as a loss, where to draw system boundaries, which components act as controllers, and what makes a control action unsafe—is where AI-specific considerations entered and where the case studies differed from one another; Section 4 details these adaptations. The lead author then cross-checked the hazards identified in the analysis against relevant literature and subject-matter experts to sanity-check the results from the STPA. The co-authors of the paper also reviewed each of the analyses and provided feedback on the application of STPA.

3.3 Developing the PHASE guideline and examining its affordances

Through reflexive analysis of the case studies and the process of applying STPA, the lead author developed Process-oriented Hazard Analysis for AI Systems (PHASE), a step-by-step guideline for conducting system-level hazard analysis of algorithmic harms in

AI systems. This work was informed by iterative discussions and feedback with the co-authors, and the joint reflexive examination of both the case study findings and the analysis process enabled the identification of the key affordances that PHASE provides for analysts. The guideline was further refined based on feedback from a pilot study, described in the supplementary material.

4 PHASE Guideline: Interpreting STPA for AI System Case Studies

Next, we present observations from applying STPA across the three case studies and highlight features unique to PHASE. Before turning to the individual steps, we briefly summarize how PHASE adapts each of the four STPA steps for AI systems.

In Step 1 (identify the purpose of the analysis), PHASE broadens how losses are conceived by drawing on value-sensitive design and by surfacing the value tensions among stakeholders, so that sociotechnical losses are considered alongside safety-critical and performance-related ones; it also draws system boundaries around the development and use processes of ML-based systems rather than around the model itself, since developers exercise control over data and design choices but not over what a model learns. In Step 2 (create a control diagram), PHASE accommodates multiple and dynamic control structures, because the way ML-based systems are used is fluid—a component can act as a controlled process in one context and as a controller in another—and it introduces components specific to AI systems, such as datasets, models, safety filters, and learning agents, as controllers or controlled processes. In Step 3 (identify unsafe control actions), PHASE summarizes the unsafe control actions characteristic of AI systems—functional failures, poor design decisions or misuse, and coordination or communication breakdowns; what distinguishes these from unsafe control actions in traditional STPA is that their unsafeness frequently depends on sociotechnical context rather than on the technical timing or sequencing of a control action. Finally, in Step 4 (identify loss scenarios), PHASE captures the loss scenarios also found in conventional STPA—missing feedback loops, incorrect mental or process models, and technical malfunctions—but adds a class unique to AI: loss scenarios arising from emerging and evolving system capabilities, where changes such as retraining, updates, or expanded use outpace existing safeguards, a source of hazard that traditional STPA, applied to systems with fixed capabilities, does not typically consider.

The PHASE guideline, along with a detailed summary of the case studies, is provided in the supplemental material.

4.1 Step 1: Identify purpose of analysis

Applying the first step of STPA - *identify purpose of analysis* - to AI systems required a substantive reinterpretation. In traditional applications of STPA, this step typically centers on defining safety-critical losses and establishing system boundaries around controllable technical components. Our case studies showed that, for AI systems, this framing is too narrow: both the notion of loss and the selection of system boundaries are deeply shaped by sociotechnical context, value tensions, and the processes through which AI systems are developed and operated. In response, PHASE adapts

the first step of STPA by broadening the interpretation of loss to incorporate insights from value-sensitive design and by emphasizing process-oriented system boundaries aligned with the algorithmic supply chain. The sections below describe our observations from the case studies and how they informed these adaptations.

Identifying losses. The losses identified for each case study go beyond the traditional definition of safety (e.g., loss of life, damage to property or environment). As shown in Table 1, the identified losses consider a broader set of emergent harms from AI systems. Our reflexive analysis reveals that there are three different categories of losses pertinent in an STPA of an ML system: 1) traditional safety-critical losses, 2) performance-related losses, 3) sociotechnical losses. *Safety-critical losses* represent losses in the traditional definition of safety, such as loss of human life, property damage, and the environment (C1.L1). *Performance-related losses*, on the other hand, emerge when the AI system does not meet performance expectations (C1.L2) or when the individuals, groups, or organizations experience loss of reputation or economic loss (C3.L6, C2.L4). Lastly, *sociotechnical losses* cover a broad range of sociotechnical harms that individuals and groups can experience that may not be safety-critical or performance-related (C1.L4, C2.L2, C3.L2). Considering the growing body of research in algorithmic harms and our findings from the case studies, we found it much more appropriate to use existing taxonomies [13, 66, 81], industry-specific regulatory requirements [1], and community-based and participatory research methods [25, 78] for loss identification of AI systems.

Defining system boundaries. Our selection of system boundaries was guided by the structure of the algorithmic supply chain and by the information available about each case, as well as by where stakeholders indicated that they could meaningfully exercise control. Rather than centering boundaries on individual models, we drew them around development and use processes, which enabled a more thorough analysis of how hazards emerge across the lifecycle of an AI system. The number of system boundaries is not prescribed by PHASE; analysts define as many as are needed to cover the loci of control relevant to the identified losses. In our cases, three boundaries per system happened to capture the accessible points of control, but this is illustrative rather than a requirement, and prior work has shown that the ML development lifecycle can be decomposed into many more stages. The boundaries also differ across cases—for example, integration or a distinct operational boundary appears in some systems but not others—because the systems, their paradigms, and the points at which developers, users, or automated agents reported having decision-making authority differ. Importantly, drawing boundaries around sub-processes does not discard the whole-system view: because boundaries are connected through shared control actions and feedback, interactions across them remain visible rather than being lost, and a hazard localized in one boundary can propagate through these links to losses defined in another.

Across all cases, we identified system boundaries within data collection and processing (e.g., C1.SB1, C3.SB1), where developers made decisions about data sources, feature inclusion, and labeling; and within model development, evaluation, and integration (e.g., C1.SB2, C2.SB1, C3.SB2), where choices about learning algorithms,

Table 1: The losses identified across the three case studies. The three types of losses (safety-critical, performance-related, and sociotechnical harms) are identified with #.

Case 1 (C1) - Early alert system	L1. Loss of life or injury to the preterm infants #safety-critical L2. Loss of time (in terms of efficiency) #performance-related L3. Loss of reputation/credibility for the physician in charge of a case #performance-related L4. Loss of patient privacy (regulation about medical data) #sociotechnicalharm
Case 2 (C2) - Insulin injection	L1. Loss of life or injury to the diabetic patient #safetycritical L2. Loss of quality of life (i.e., comfort and autonomy) #sociotechnicalharm L3. Loss of efficiency #performance-related L4. Loss of money #performance-related
Case 3 (C3) - Story boarding	L1. Loss of creativity #sociotechnicalharm L2. Loss of diversity #sociotechnicalharm L3. Loss of accessibility #sociotechnicalharm L4. Loss of efficiency #performance-related L5. Loss of quality #performance-related L6. Loss of reputation #performance-related

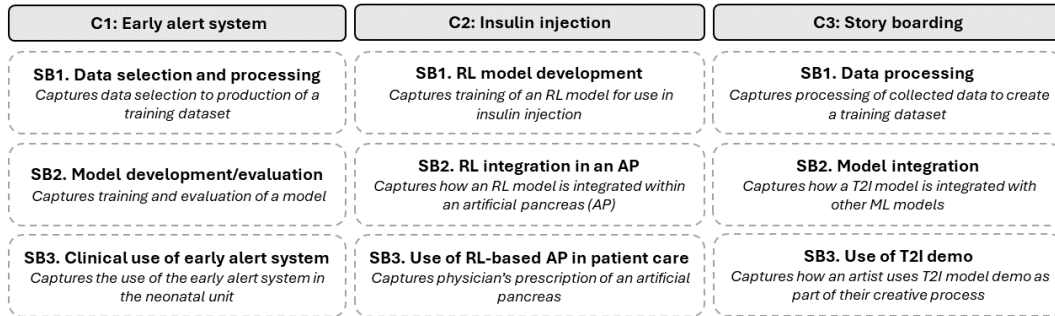


Figure 2: Three system boundaries (SB) defined for each of the case studies (C). In practice, analysts are free to define SBs following Section 2.3.1. The items under the Case title describe the scope of the SB we defined for the Case Study.

architectures, metrics, and outputs were exercised. However, despite this apparent control, stakeholders consistently noted limited visibility into—and control over—the learned latent representations of ML models, which constrained the usefulness of drawing system boundaries narrowly around the model itself.

The contrast between the reinforcement learning (RL) case and systems built around opaque, black-box generative models further illustrates this point. In the RL case, we identified an operational system boundary (C2.SB2) where control shifted from human operators to the algorithmic agent, which autonomously adjusted the amount of insulin delivered to a diabetic patient. This clear locus of control during operation made it possible to reason explicitly about hazards arising from the system’s autonomous decision-making. In contrast, for opaque generative systems such as text-to-image models, stakeholders reported limited ability to interpret or directly control model behavior at inference time. For a general-purpose system such as a T2I platform, the space of possible uses is open-ended, which makes boundary definition especially consequential: a boundary drawn loosely around “use of the demo” is too vague to determine whether a given hazard is controllable within it. We therefore drew the downstream boundary narrowly around the generation and dissemination of a specific output (e.g., a storyboard

used for video creation, C3.SB3), where a hazard such as the dissemination of false or harmful content becomes controllable—through the prompts entered, the safety filters applied, and the decision to publish—rather than around the platform’s use in the abstract. Consequently, system boundaries for these systems were more meaningfully drawn around upstream development processes and downstream use contexts—such as content generation and dissemination (e.g., C3.SB3) - where human actors retained control over how outputs were produced, interpreted, and shared.

Outlining Hazards. Hazard identification largely followed established STPA practice and was driven by the system boundaries and loss definitions established in the first step of the analysis. The hazards identified across the case studies aligned with the three loss categories described above and often spanned multiple losses. For example, the hazard “The system creates a model or output that does not meet performance requirements” for C1.SB2 could degrade an early-warning system and, in extreme cases, contribute to loss of life (C1.L1). Other hazards reflected well-documented sociotechnical concerns in the Fairness, Accountability, Transparency and Ethics (FATE) and RAI literature, such as lack of interpretability (C1.SB2; L1, L3) [12] and the dissemination of false or harmful

content (C3.SB3; L5, L6) [14]. Here, the input and output safety filters referenced throughout the T2I case are classifiers that screen prompts and generated images to block disallowed content; their performance refers to how accurately they do so, that is, their false-positive rate (blocking benign content) and false-negative rate (allowing harmful content through), and a poorly designed filter is one whose threshold admits too many harmful outputs or suppresses too much legitimate use.

4.2 Step 2: Create a control diagram

A key challenge in applying STPA to AI systems is that some components do not occupy fixed roles as controllers or controlled processes. For example, an AI assistant may function as a controlled process in one context (e.g., supporting human decision-making) and effectively act as a controller in another when users are organizationally required to follow its outputs. In PHASE, we explicitly account for this role fluidity by encouraging analysts to model control relationships as context-dependent rather than static. Capturing this role fluidity at the control-diagram stage is essential, as hazards depend not only on system behavior but also on how control shifts between humans and AI systems in practice.

Using the core concepts of STPA, we mapped controllers, controlled processes, control actions, feedback, inputs, outputs, and control hierarchies within each selected system boundary. Across the case studies, we were able to translate these concepts to AI systems and interpret how they manifest across both development and operational contexts. We constructed a control diagram for each system boundary, yielding a total of 9 diagrams; the full set is provided in the supplementary material, and we show one boundary per case study in Figure 3 to illustrate the range of control structures across the three paradigms. To read such a diagram, one traces each control action downward—the constraints a controller imposes on the process below it—and each feedback signal upward—the information returned to the controller—so that, together, they reveal where control is adequate and where it may break down. Figure 3 presents one representative control diagram from each case study. The elements represented as controllers or controlled processes ranged from individuals and teams to technical artifacts. In many cases, controllers were individuals or groups responsible for decision-making during development or use. For example, in C1.SB1, controllers included the data collection team, data processing team, and hospital database manager, while the collected dataset functioned as a controlled process. Similarly, in C3.SB2, technical artifacts such as input and output safety filters and text-to-image models were modeled as controlled processes. However, individuals and teams were not always the primary controllers. In some contexts, control was exercised by technical artifacts themselves. For instance, in C2.SB3, the reinforcement learning agent and the insulin pump acted as controllers by autonomously determining insulin delivery. In other cases, control relationships depended on organizational and use-context constraints. For example, in C1.SB3, when use of an early alert system was mandated for physicians, the AI system constrained clinical decision-making, effectively positioning it as a controller despite its nominal role as a decision-support tool. These examples illustrate why treating AI systems as having fixed roles obscures important sources of risk.

The top-to-bottom representation of the control hierarchy (with higher levels indicating greater control) was generally effective for visualizing these relationships, but the ordering of the hierarchy was often context-dependent and subject to change based on system design choices, organizational policies, or patterns of use. Across the case studies, control actions ranged from development-time decisions (e.g., dataset processing and model design choices), to use-time actions (e.g., prompts entered by ML artists interacting with a text-to-image system), to technical constraints (e.g., threshold settings for safety filters). Feedback took the form of evaluation metrics, sensor outputs, user feedback, or other forms of human communication. Inputs and outputs captured both technical artifacts (e.g., training datasets) and coordination among individuals and teams, such as collaboration between physicians and developers to ensure clinically appropriate model behavior (C1).

4.3 Step 3: Identify unsafe control actions

Overall, we found that Step 3 of STPA—identifying unsafe control actions (UCAs)—could be applied to AI systems with minimal modification. The core STPA logic of examining control actions in relation to system-level hazards was transferred directly; however, careful interpretation of the four conditions under which a control action becomes unsafe was essential in AI contexts. In particular, whether a control action is provided, not provided, provided too early or too late, or applied too long or too briefly often depended on sociotechnical context rather than purely technical timing or sequencing. From our analysis, we identified three main trends in UCAs applicable to AI systems.

Functional UCAs The most common type involved *functional UCAs*, where inadequate control resulted from a malfunction or dysfunction of the controller. For example, in C2.SB2, the insulin pump in the artificial pancreas failed to release the required amount of insulin at a given time. Similarly, in C1.SB3, an incorrect alarm from the early alert system could negatively influence a physician’s decision-making. These UCAs were primarily associated with performance-related and safety-critical losses.

UCAs from poor design decision-making or misuse A second type of UCAs stemmed from *poor design decisions or misuse* by experts, system developers, or users. For instance, in C3.SB2, model developers could select an inappropriate safety filter threshold, enabling harmful prompts or image outputs. In C1.SB1, the data processing team could select incorrect clinical variables as features, leading to misleading model behavior. These UCAs highlight how design-time decisions can render otherwise correct control actions unsafe in context.

UCAs from communication/coordination The final type of UCAs involved *inadequate organizational coordination or miscommunication*. For example, model developers might receive incomplete or incorrect safety requirements, or lack sufficient time to address them before deployment deadlines (C3.SB2). In these cases, unsafe control actions emerged not from technical failure, but from misaligned processes and constraints across teams.

4.4 Step 4: Identify loss scenarios

The original STPA process was highly effective in the process of understanding loss scenarios for AI systems. We brainstormed a

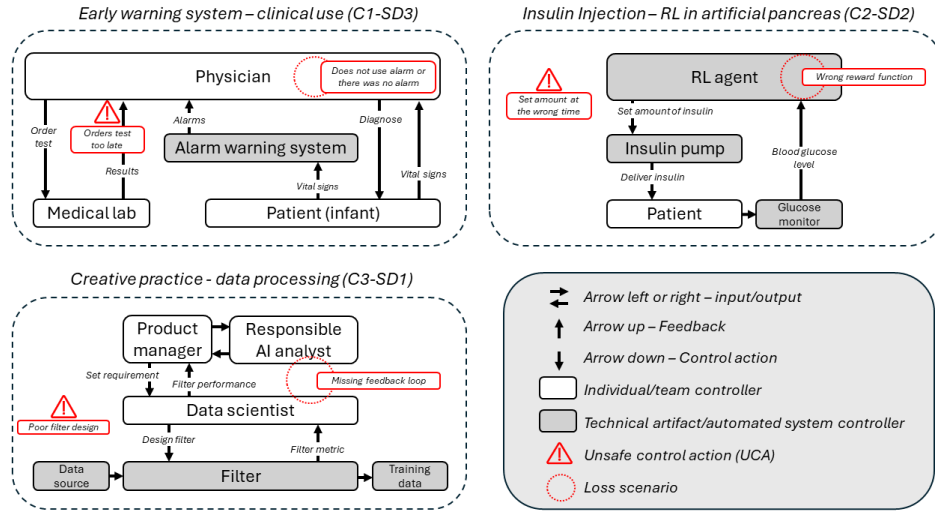


Figure 3: This diagram illustrates the simplified control structure. Each of the System Diagrams (SD) corresponds to one System Boundary (SB). We have included only three of the system boundaries in this diagram, one from each case study.

list of loss scenarios by thinking through all the elements of the control diagram. A diverse range of loss scenarios were identified, which included organizational/institutional causal scenarios (i.e., poor understanding or communication), interaction issues (e.g., incorrect mental model) and technical challenges (e.g., missing metrics/evaluations of a system). Across the case studies, however, we observed a recurring class of loss scenarios unique to AI systems: losses arising from emerging and evolving system capabilities. These scenarios did not necessarily stem from immediate component failure, but from changes in model behavior over time—such as updates, retraining, or expanded use—that outpaced existing assumptions, safeguards, or organizational processes. In addition, we observed two other prevalent forms of loss scenarios. A common institutional causal scenario across our case studies involved missing feedback loops between different teams that are in charge of various parts of the development process. For instance, often model development teams use pre-existing datasets that have been collected and processed by another group and at a different time frame than when the model (e.g., T2I model) is being developed. In such cases, there is a missing feedback loop between the developers of the model and creators of the dataset (e.g., C3-SB2). Another type of causal scenario is when the user has an incorrect mental model of the AI system, meaning that they do not understand how to use it correctly or how it works. This scenario is characteristic of AI systems deployed to non-expert users: unlike traditional safety-critical software operated by trained specialists, AI tools are increasingly used by people without an accurate model of the system’s competence or limits, and PHASE captures this through a mismatch between the controller’s process model and the actual behavior of the controlled process. For instance, for C1.SB3, if the physician does not understand the reason for the alarm, they could make the wrong lab order or diagnosis. The third type of loss scenario is *technical challenges* which occurs when a component (i.e., a pump, ML model, safety classifier) is functioning erroneously. This could lead to inadequate control when making decisions about a patient’s

health. The type of loss scenarios we identified is not exhaustive. However, they provide an understanding of the type of causal loss scenarios for inadequate control in AI systems.

5 Affordances of System Theoretic Perspective for algorithmic harm from AI systems

Based on our reflexive analysis, we identified four affordances observed across our case studies that build on STPA when applied to AI systems using our PHASE guideline. We stress that the value here is not system-level analysis in itself—which STPA already provides—but the specific adaptations PHASE makes for AI: an expanded loss taxonomy that includes sociotechnical harms, the treatment of AI components as context-dependent controllers or controlled processes rather than fixed-role artifacts, and loss scenarios that account for evolving model capabilities. The affordances described below follow from these adaptations, and it is these adaptations, rather than the general system-centric stance, that we claim as PHASE’s contribution.

5.1 Affordance 1: Analyze AI at the systems level

Since STPA is designed to analyze a process or an artifact at the systems level, PHASE inherits this affordance. Prior work suggests that algorithmic harms from AI arise from system-wide issues [64, 68, 73]. For instance, representational harms are system-level issues that “occur when algorithmic systems reinforce the subordination of social groups along the lines of identity, such as disability, gender, race and ethnicity, religion, and sexuality” [34, 66]. They arise from a combination of ad-hoc factors such as inaccurate ML model performance for minority groups, misuse of the system in a given application, or exclusionary social norms. Instead of narrowly focusing on “fixing” an unbalanced training dataset, system-level analysis prompts us to examine how this dataset will interact with

other parts of the system, such dataset analysts, other ML models, or the user interface [7, 77].

STPA is a systematic method that establishes possible causal chains between undesirable outcomes of the system and inadequate control. As such, the *losses* and *harms* identified in the first step of PHASE (Section 4.1) define the negative impact of the system an analyst aims to prevent or mitigate. Subsequently, UCAs (Section 4.3) explicitly defines all possible cases of inadequate control in the chosen system boundary that lead to a hazard. Loss scenarios (Section 4.4) finally establish the causal link connecting the undesirable outcomes to specific unsafe states of the system. For example, we identified the hazard “*H: Clinician misdiagnoses the patient.*” to stem from *inadequate control* over the proper use of the early alarm system and timing for ordering a blood test, which can lead to *L: loss of life*. In this case, the control action of “*clinical use of the early alert system*” could be unsafe and in several loss scenarios, such as when the clinician does not understand how to use the early alert system. Notably, this hazard would not surface from a component-level analysis of the model or dataset alone: a fairness or accuracy audit of the alert model in isolation could report acceptable performance, yet the loss still arises from the interaction between an imperfect alert and a clinician who over-trusts or misreads it. It is only by analyzing the dataset, model, and user together that this system-level hazard becomes visible.

5.2 Affordance 2: Accounting for social factors and sociotechnical harms

Formalizing losses at the beginning of STPA mobilizes a discussion and instantiates the need for consensus amongst stakeholders about harms that must be prioritized. Amongst practitioners and scholars, there is a divergence of perspectives on the types of harm that need to be assessed and mitigated for AI systems [32, 38, 62]. Currently, company policies and ad-hoc conversations amongst limited, invited individuals shape practitioners’ choices on what harms need to be minimized or fully prevented [2]. However, following PHASE structures the analysis so that a broad set of sociotechnical harms is considered at the outset. It cannot compel an analyst to be exhaustive; rather, by making the enumeration of losses the explicit first step, it renders the omission of such harms visible and open to challenge rather than leaving it implicit. Furthermore, in defining the system boundary, PHASE connects the losses to a network of individuals, teams, technical artifacts, automated systems, and institutional mechanisms which form a system boundary along the AI development cycle as opposed to just examining a loss by focusing on one of these elements (i.e. blaming a developer or assuming ML models cannot be explained). Lastly, the control diagram and its components in PHASE provide the conceptual flexibility to map social and technical elements within a system boundary. For example, a controller could be a human (e.g., an artist) who has a specific mental model of how to use a T2I platform. It could also be an RL agent with a map of the environment (*process model*) and a reward model (*control algorithm*) that guides the suggested dosage of insulin (*control actions*). This conceptual flexibility allows for the integration and prioritization of sociotechnical factors in AI harms analysis.

5.3 Affordance 3: Identify accountability chain by establishing traceability

Traceability in harm reduction is a pressing concern in the AI community. Recent studies indicate that responsible AI practitioners are frustrated by the lack of systematic methods that establish a connection between a specific set of harms and the means to mitigate the harm [46, 57, 67]. This problem is exacerbated by the fact that many systems are developed by a multitude of individuals and groups across institutions that hardly interact directly with one another [51, 83]. That is, many unsafe system states can accumulate between individual decision-makers, the technical system, and the development process without the situation being detected or managed [37, 83]. As described earlier, the STPA framework allows us to connect harms to inadequate control within a system. Furthermore, it allows an analyst to make explicit who has control over what, and make the harms and hazardous states of a system traceable for everyone. Specifically, the concept of hierarchical control structure as interpreted in PHASE engages the analyst to map who or what has control in relation to other actors or artifacts by mapping their interaction using control action and feedback. For instance, by drawing the control structure, we needed to examine whether the clinician had control over the early alert system meaning that they could choose to use the warning upon reception (i.e., feedback) or that they needed to use the warning from the early alert system (i.e., the control action) by thinking through how the early warning system is integrated into clinical practice. Understanding the control hierarchy fosters an understanding of how power, control, and responsibility are distributed in a system, which ultimately results in a more managed chain of accountability for responsible AI deployment. Concretely, this is what lets an analyst localize a harm to its source. If the early alert system contributes to a harmful misdiagnosis, tracing the loss scenario back through the control structure distinguishes whether the inadequate control originated in data collection (for example, a context-specific perturbation missing from the training data, surfaced as a missing feedback loop between the data and model teams), in the model (a training deficiency, surfaced as an inadequate control algorithm), or in use (a clinician who does not understand the alert, surfaced as an incorrect process model). Because a single loss scenario can implicate more than one control action or feedback link at once, the same structure also makes explicit when a harm is combinatorial—arising from the joint effect of, say, a data gap and a user’s misunderstanding—rather than attributable to any one component in isolation.

5.4 Affordance 4: Monitor and address emergent hazards over time

STPA framework enables an analyst to monitor and account for changes in individuals, teams, technical artifacts, and institutional mechanisms over time. By emergent hazards over time, we mean hazards that were absent or not analyzable at initial design but arise as the model, its use, or its organizational context changes—for example, as a reinforcement learning agent adapts its policy during operation or as a generative model is retrained or updated after deployment. As C2 illustrates, this includes the changes that stem from the adaptivity of AI systems such as RL. These adaptations can be reflected in various components of a control diagram, such

as control actions, feedback loops, and elements of a controller (e.g., decision-making protocol, process/mental model of the world). For instance, the change in the output of an AI model could be captured in a control action as constraints/information a clinician or an artist receives. Furthermore, changes in an AI model's latent representation could be captured in the process model of the controller or controlled process (depending on whether a model is acting as a controller or is a controlled process). Changes in how individuals, teams, and organizations perceive and operate, such as changes in how an AI model is evaluated, could be captured as part of the feedback loop between the model and the developer team. Such concrete means of accounting for changes in the system allow analysts to monitor the dynamics systematically and iteratively, prompting them to examine emergent hazards as AI systems change in how they are integrated or interact with their sociotechnical context. This is a critical affordance given the rapid pace of AI development and increased interest in adaptive and agentic AI systems [65, 80].

6 Discussion

In this paper, we employed a case study approach to translate STPA for the analysis of AI systems. We illustrated that STPA framework can not only be translated to analyze AI systems and identify potential sources of algorithmic harm, but that the PHASE implementation of the analysis enables four affordances critically missing in existing AI assessment frameworks. STPA framework allows analysts to map out sociotechnical system dynamics, and instantiate accountability by examining harms at the system level and tracing them to control methods for harm reduction. Moreover, the framework allows for accounting of changes in individuals, collectives, AI models and artifacts, and institutional processes that could happen over time. Here, we discuss the broader implications of adopting a system-theoretic perspective for AI governance and building a responsible AI culture in organizations. We also elaborate on limitations and avenues for future work.

6.1 Implications for policy and AI governance

From self-governance mechanisms in corporations to national regulations on AI, there has been a growing effort to practice due diligence and establish accountability mechanisms across the AI industry [1, 72]. As illustrated in our findings (Section 5.3), establishing traceability between harms and causal scenarios can be a powerful tool toward this end. It can enable an analyst to define a division of accountability where specific roles/groups are held responsible for maintaining adequate/safe control of certain subsystems [40]. Although our system boundaries did not explicitly focus on internal or external governance mechanisms, some of the controls added by internal governance mechanisms were observed in our case studies. For example, for *C3-SB1* as captured in Figure 3, the responsible AI analyst provides input on safety requirements to product managers about datasets, which are then passed onto the data scientists. These safety requirements would typically be developed based on existing regulations or internal company policies [2]. Given that STPA has already been used in other domains to examine and improve governance mechanisms; we expect PHASE to be able to afford the same function for AI systems [39].

For example, such a process could point out the missing feedback loop between the data scientist and the responsible AI analyst in *C3-SB1*, which hints at the lack of quality control of the training dataset and the potential need for a data quality approval mechanism. In addition to improving internal accountability protocols for a company, external actors such as standard organizations could leverage these affordances to establish best practices for conducting hazard analysis, and mobilize policymakers to refer to these standards in the regulation for certain AI products [70].

6.2 Building a safety culture: systems lens for responsible and safe AI

Given the focus of our three case studies in the product development context, we also observed how existing culture within the organization and management practices can contribute to sociotechnical harm. For example, in *C3-SB1*, the loss scenarios for the UCA of poor safety filter design could stem from a responsible AI analyst not having enough time to review the dataset and provide safety requirements to the product manager before the product needs to be launched. The presence of toxic or profit-driven culture within the tech industry and its relationship to the quality of products deployed in the market is not new [50]. PHASE helps account for this social factor and its relationship to a specific set of harms. Our hope is that such explicit acknowledgment of harms that can stem from social contexts can help organizations prioritize the implementation of a safety culture. Even though STPA could be a tool for building a safety culture, adoption of STPA demands a company culture that values safety and is willing to allocate resources to it. Lack of incentives and financial/human resources limits the applicability and effectiveness of many of the existing responsible AI/AI ethics frameworks [46, 57]. The persistence of these challenges for STPA would hinder adoption even with the development of guidelines that translate and illustrate the framework for RAI practice.

6.3 Limitations and future work

Recent discussion about the possible risks of developing agentic systems suggests that agentic AI will be inherently dangerous considering that lack of any meaningful methods to control these systems and comprehensively understand their process model of the world [21, 61, 65]. While none of our case studies involve an agentic AI, it was clear that the PHASE effectively maps where and what humans and technical systems/processes *have or do not have* control. We believe that this mapping can offer valuable insights in contextually examining increasingly agentic systems, especially in investigating areas of inadequate or lack of control to examine systems that are inherently unsafe or uncontrollable.

There are many known limitations of STPA that also transfer to PHASE. First, STPA frames interaction between processes or agents as control and feedback. Such framing limits how nuances in human relationships can be captured, such as collaborative decision-making and social influence between agents. Furthermore, similar to any analytical process, analysts must make normative choices, assumptions, and simplifications in conducting PHASE. This means that the views, values, and perspectives of the analysts affect the final analysis. Relatedly, the conceptual flexibility that lets analysts

map social and technical elements within a boundary is also a limitation: analysts may equally choose not to include such elements, so the same flexibility that is a strength for diverse use cases can weaken an analysis in less careful hands. To improve robustness against this analyst-dependence, PHASE can be conducted by multiple or deliberately diverse analysts and the resulting hazard and loss-scenario lists cross-validated against one another. Therefore, it is imperative to have well-established industry standards on what a quality PHASE should look like. We provide a running example in the supplementary material as a starting point. A further limitation concerns generality: our findings rest on three case studies, and although these were chosen to span distinct ML paradigms and interaction modes, they cannot represent the full AI landscape, so broader application is needed to test how far these adaptations generalize. Lastly, PHASE is a comprehensive hazard analysis that requires adequate expertise and time; to lower this barrier, the supplementary guideline provides step-by-step instructions, templates, and a worked example, though effective use still assumes basic familiarity with the system being analyzed and with control-based reasoning.

7 Conclusion

In this work, we interpret and translate STPA, a well-established system safety hazard analysis framework. We find that system safety models and methods offer a complementary perspective to existing responsible AI efforts. It helps us to defining safety as an emergent property of the system. Through three real-life case studies, we conclude that the concepts and steps of STPA apply readily to the analysis of AI systems. We developed the PHASE guideline that translates these concepts for the analysis of AI systems. Pilot testing the guidelines with students in a graduate-level responsible AI course helped us validate and improve the final PHASE guideline provided in the supplementary material. We find that PHASE affords an analyst the ability to adopt a sociotechnical lens, establish traceability, and account for changes in a system over time. We invite the AI community to leverage these affordances to address growing safety concerns with rapidly changing AI systems, establish traceable AI governance practices, and improve safety culture in the technology industry for the safety of all.

8 Generative AI Usage Statement

The first author used ChatGPT (version 5.2) sparingly to obtain suggestions related to grammar, writing style, and conciseness. All suggestions were reviewed critically, and none were incorporated verbatim. Grammarly was additionally used for grammar and spelling edits. The remaining authors primarily provided feedback through verbal discussions and written comments on the manuscript and did not use generative AI tools for providing feedback.

9 Ethical Considerations Statement

In this study, we collected information at multiple stages of our methodology as described in Section 3 of the paper. In the information gathering stage for Case Study 1 and Case Study 2, we primarily relied on publicly available resources and consulted subject-matter experts as needed to clarify our understanding of the projects. As such, and following our institution's research ethics board policy,

research ethics approval for the two cases was not needed. Case Study 3, on the other hand, involved a qualitative study where we recruited and conducted semi-structured interviews with relevant stakeholders along the development process. As this pertains to designed interventions with collection from human participants, we followed and received the necessary institutional approvals prior to conducting the study. All the participants (15 for the workshop and 8 for interviews) for Case Study 3 provided informed consent.

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A Supplementary Material

Process-Oriented Hazard Analysis for AI Systems

Overview

This document details a set of guidelines for applying System Theoretic Process Analysis (STPA)^{1,2} for algorithmic systems that are based on machine learning (ML) algorithms. We call this Process-oriented Hazard Analysis for AI Systems (PHASE). STPA is a well-established analytical framework from the field of safety engineering developed by Nancy Leveson. It takes a control and system theoretic approach to identify key losses and hazards to be avoided and examines how elements of a system and their interactions relate to the given losses and hazards (Leveson 2016). Traditionally, STPA was developed for complex safety-critical systems whose failure could result in loss of life, significant property damage, or damage to the environment (Knight 2002)³. PHASE illustrates a way of interpreting STPA concepts and steps for ML-based algorithmic systems, recognizing that many of the harms from deployment of these systems go beyond traditional safety-critical losses such as loss of life.



Figure 1. Flow of analysis for PHASE is modelled after the four key steps in STPA

What is in this guideline?

PHASE elaborates on the four stages of the STPA framework, which broadly include 1) Identifying losses and hazards, 2) Mapping out system dynamics, 3) Examining unsafe interactions, and 4) Identifying potential causal scenarios for these unsafe interactions. Next, we discuss the reasons for creating this guideline and how it was developed. We also discuss remarks about the mechanics (i.e.,

¹ Leveson, N., & Thompson, J. (2018). *STPA Handbook*.

https://psas.scripts.mit.edu/home/get_file.php?name=STPA_handbook.pdf

² Leveson, N. (2011). *Engineering a Safer World: Systems Thinking Applied to Safety*. MIT Press.

³ Knight, J. C. (2002). Safety critical systems: challenges and directions. *Proceedings of the 24th International Conference on Software Engineering. ICSE 2002*, 547–550.

who? when?) of applying PHASE. The remainder of this document describes the interpretation of these steps formally (with an accompanying example for the [COMPAS](#) algorithm).

Why did we create PHASE?

1) Bringing a Control-Theoretic Perspective to ML Systems

Machine learning (ML) systems are inherently probabilistic, yet they operate within structured interactions where different sub-systems influence each other. PHASE applies control theory to ML-based systems, framing them as networks of interconnected components—such as models, datasets, and human actors—that either control or are controlled by other sub-systems. Even though ML models are probabilistic, this perspective helps us analyze how different components affect system behavior.

This guideline provides a step-by-step approach to applying the Systems-Theoretic Process Analysis (STPA) framework to ML-based systems, emphasizing their complexity and the interactions between automated, algorithmic, and human sub-systems. Additionally, it highlights the evolving nature of ML systems, which change over time based on factors such as input data, model architecture, and usage patterns. PHASE equips practitioners with the necessary analytical framing to account for these dynamics.

2) Addressing a Broader Range of Harms Beyond Safety-Critical Failures

Many algorithmic systems operate in safety-critical contexts where failures can result in physical harm or damage to property and the environment. However, the risks of ML-based systems extend beyond traditional safety concerns. In recent years, media reports and academic research have documented various adverse impacts, including discrimination against marginalized groups, the spread of misinformation, and privacy violations.

While existing STPA applications primarily focus on safety-critical domains, the STPA handbook acknowledges the framework's potential for analyzing a wider range of harms. PHASE serves as a guideline for applying STPA to ML-based systems in ways that go beyond traditional safety risks, enabling a more comprehensive hazard analysis that accounts for social and ethical concerns.

3) Making STPA Accessible to ML Practitioners

The original terminology and concepts in STPA can be difficult to navigate for ML practitioners such as product managers, responsible AI analysts, and ML engineers/researchers [ref]. PHASE bridges this gap by translating STPA concepts into a more accessible framework, ensuring that practitioners can effectively understand and apply the methodology in their work.

By providing this structured guideline, PHASE makes it easier for ML professionals to conduct hazard analyses, assess risks, and design more accountable and reliable ML-based systems.

How did we develop PHASE?

To develop PHASE, we followed a structured, iterative process that combined industry insights, comparative analysis, and real-world case studies. Here are the key steps we took:

- 1) **Understanding Industry Practices and Practitioner Perspectives**

We began by examining current ethical and social risk management practices used in the industry. We also gathered insights from ML practitioners about their experiences with safety engineering frameworks such as Failure Modes and Effects Analysis (FMEA) and Systems-Theoretic Process Analysis (STPA). Practitioners particularly valued STPA's focus on starting with harms and its ability to map out complex system dynamics.

- 2) **Comparative Analysis of FMEA and STPA for Text-to-Image Models**

We conducted a comparative analysis of FMEA and STPA in the context of text-to-image models used in creative practice. This analysis highlighted that STPA offers a more structured approach to identifying potential sources of harm and accounts for hazards arising from contextual factors, such as how a company structures its development processes and how an ML system is embedded in a product. We followed the STPA methodology as outlined in the Leveson and Thomas handbook.

- 3) **Applying STPA to Additional Case Studies**

Beyond text-to-image models, we applied STPA to two additional case studies:

- a) The use of predictive algorithms in clinical decision-making.
- b) The use of reinforcement learning (RL) algorithms for determining injection dosages in artificial pancreas systems.

These applications helped refine our understanding of STPA's strengths and informed the first iteration of PHASE.

- 4) **Piloting PHASE Through Teaching and Workshops**

We tested the first version of PHASE by teaching it to students in a graduate-level course on the Ethics of Intelligent Systems. We also introduced it in workshops and tutorials, including the Sociotechnical Safety Workshop at HAII Stanford and FAccT 2024. These pilots provided valuable feedback that helped us further refine the framework.

Through this process, PHASE evolved into a structured guideline tailored for ML practitioners, bridging the gap between safety engineering principles and the unique challenges of ML-based systems.

How should you use this guideline document?

We recommend that you use PHASE and the supplemental case studies along with established STPA resources such as the STPA Handbook and instructional videos provided by the creators of STPA.

[Maybe add something about setting up PHASE on GitHub as an open-source guideline]

This type of hazard analysis can and should ideally be conducted early in developing an ML-based product such that discoveries from the process could inform future design requirements and decisions. Considering the changing nature of ML-based products, it is important to update the analysis, especially if there are changes in the control structure diagram.

The individuals who conduct this analysis need to have (or be able to gain) a global understanding of how the ML system works and who is involved in developing/launching it. Having a team of interdisciplinary experts trained in the PHASE guideline and STPA framework collaborate on the analysis is highly beneficial. Their diverse perspectives enhance every step of the process, from identifying losses and defining system boundaries to mapping the system and uncovering potential causal scenarios.

Step 1: Identify the purpose of the analysis

Identify losses and hazards

PHASE framework draws attention to the social context, the adverse impact of an ML-based algorithmic system, and potential sources of harm to identify the purpose of analysis. To identify the purpose of analysis, follow the key definitions and steps to 1) Identify losses and 2) Identify system-level hazards.

Identify losses

STPA starts with identifying *losses*, which is defined broadly as the loss of something stakeholders consider valuable. For instance, the loss of a person's life would be important in the case of analyzing a car as a system. To identify losses, start by *identifying the key stakeholders and understanding their values*. However, there is an extensive range of possible stakeholder values, and all value losses are not necessarily problematic. To develop and refine the list of losses:

- Examine if any of the common safety-critical losses are applicable to the case. These include loss of human life or human injury, property damage, and the environment. If so, enlist them as potential losses.
- Examine if performance-related losses are applicable here. Performance-related losses emerge when the ML system does not meet commonly used and desired performance metrics such as accuracy, recall, and AUC (area under the curve) value. For ML systems, performance losses could include loss of user engagement, loss of credibility, loss of public trust, and loss of venture capital.⁴

⁴ This list of performance-related losses is only illustrative, and it is not comprehensive across all ML applications.

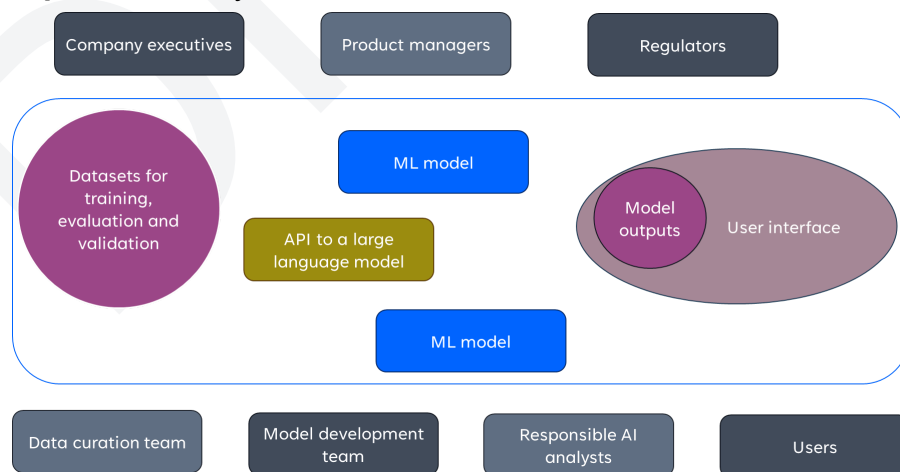
- Examine losses from ML-based systems that are not covered by safety-critical and performance-based losses. As mentioned earlier, harms from ML-based systems expand beyond inaccurate outputs and physical injury. Sociotechnical harms from algorithmic systems occur when the use of these systems leads to losses such as loss of justice, loss of the right to privacy, and loss of sense of identity. To identify such losses:
 - Use existing credible algorithmic harm taxonomies as a method of identifying potential losses. For example, the taxonomy of sociotechnical harms by Shelby et. al, outlines five categories of harms, including representational, allocative, quality-of-service, interpersonal, and social system harms (Shelby et al. 2023)⁵.
 - Re-examine and integrate the stakeholder values (identified earlier) that are critical to include, as loss of such values could result in harm to impacted individuals and groups
 - Review and monitor existing regulatory requirements for losses that are important to minimize/avoid by regulatory standards.

The final list of losses generally contains 4-6 items, and it should not exceed 10 losses.

Identify system-level hazards

Once losses are identified, the next step is to identify hazards. Before outlining the necessary steps to identify hazards, here are some key terms to help you with successful hazard identification (Leveson 2016):

- **System definition:** A system is a set of components that act together as a whole to achieve some common goal, objective, or end. A system may contain subsystems and may also be part of a larger system.
- **Hazard definition:** A hazard is a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss.
- **Hazard specification = system and unsafe condition and link to losses**



⁵ Shelby, R., Rismani, S., Henne, K., Moon, A., Rostamzadeh, N., Nicholas, P., Yilla-Akbari, N. 'mah, Gallegos, J., Smart, A., Garcia, E., & Virk, G. (2023). Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction. *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*, 723–741.

Figure 2. ML-based systems are complex and consist of many algorithmic, automated, and human sub-systems. This figure shows some of the sub-systems (technical and human) involved in ML development.

To identify system-level hazards:

- Identify system boundaries: This also defines the scope of the analysis. Select the system boundary based on three factors:
 - The boundary should include the parts of the system that its designers have some control over
 - The amount of information you have about parts of the system as the analyst
 - The stated/necessary goal of the analysis based on existing regulations/policies

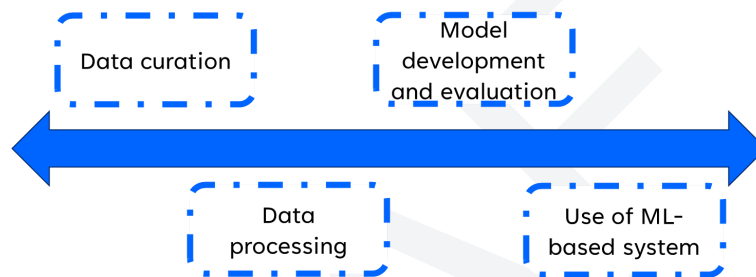


Figure 3. Examples of potential system boundaries along the ML development pipeline

For algorithmic systems, the part of the ML development process that system designers have control over includes data processing protocols for curating train/validation/test datasets, evaluation protocols for the model and its output, model architecture development, infrastructure for setting up necessary compute resources and training process, integration of different ML models in a given product, and user interaction design affordances. Once a system boundary is identified,

- Brainstorm about the hazards based on two pieces of information: the losses and the overall objective of the chosen system. Consider what system state and conditions would lead to loss X, given the objective of the system. For example, if we are looking at the scope of data processing, the objective of the system is to create a training dataset. For loss of privacy, a potential system-level hazard would be: the training dataset is not anonymized, or the training dataset breaches privacy guidelines as outlined by a specific privacy guideline. For loss of justice, a potential system-level hazard would be the training dataset only contains data from a specific stakeholder group OR the training dataset is not diverse in representation.

A list of 2-6 system-level hazards is a good place to start. Hazards need to be connected to the list of losses created in the first part of this analysis.

Example:**Losses:**

- L1: loss of reputation (for Northpointe) - (performance related)
- L2: loss of justice (representational and allocative harms)
- L3: loss of credibility for judge (interpersonal harm)
- L4: loss of privacy (interpersonal harm)

System boundary:

- S1: Use of COMPAS in a criminal trial by a judge
- S2: Data curation process for getting the training and evaluation datasets for the algorithm
- S3: Model development process for COMPAS

Hazards:

- For S1:
 - H1: The judge does not understand or misunderstands the prediction. (L2, L3)
 - H2: The judge misuses the prediction by COMPAS. (L2, L3)
 - H3: The judge misuses the system.
- For S2:
 - H1: The curated data is not anonymized (appropriately) (L1, L4)
 - H2: The curated data is low-quality in terms of validity/correctness (L1, L2)
 - H3: The curated data does not cover specific racial groups. (L2, L3).
- For S3:
 - H1: The model cannot predict recidivism accurately (L1, L2, L3).
 - H2: The developers do not understand how the model makes its predictions (L1, L2).

Step 2: Create a control structure diagram

Mapping system dynamics

The second step involves drawing a diagram that shows the sub-systems within our system boundary and their connections. This diagram is called a control structure, and Leveson and Johnson define the key terms of the control structure as:

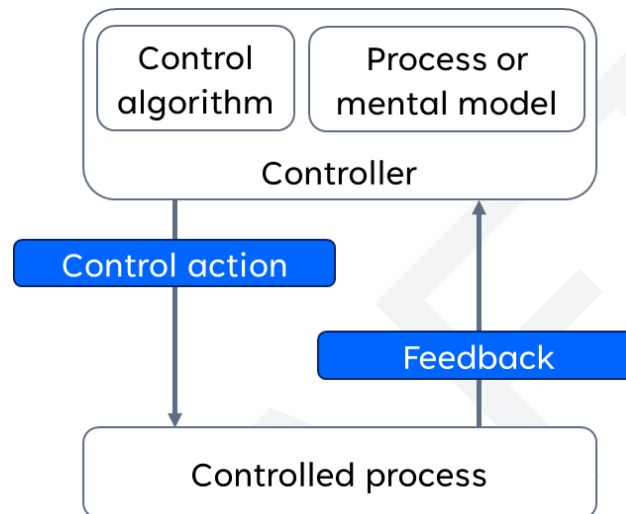


Figure 4. A simple control structure diagram with one controller and one controlled process

- **Control structure:** A hierarchical control structure is a system model that is composed of feedback control loops. An effective control structure will enforce constraints on the behavior of the overall system.
- A hierarchical control structure is composed of control loops. In general, a controller may provide **control actions** to control some processes and to enforce constraints on the behavior of the controlled process. A controller may be represented by a human decision-maker, automatic decision-making, or a combination thereof. Various constellations of human and automatic control exist, often expressed in terms of ‘level of automation’. The **control algorithm** represents the controller’s decision-making process—it determines the control actions to provide based on some inputs (information or a sensor signal).
- Controllers also have **process models** that represent the controller’s internal beliefs used to make decisions. Process models may include beliefs about the process being controlled or other relevant aspects of the system or the environment. In a formal sense, they specify four conditions:
 - Goal condition: the objective and constraints of the controller
 - Action condition: the actions the controller has at its disposal to steer the controlled process to the objective while respecting the constraints

- Observability condition: the real-time information the controller needs about the controlled process in order to determine the right course of action
- Model condition: the broader understanding of the controlled process, as well as how the control structure acts on it
- **Mental models** are equivalent to process models for controllers that are executed by individuals or groups of people. If the control structure entails both human and automatic controllers, the human controller's mental model also includes the beliefs and understanding about the automatic controller and how it affects decision-making and the controlled process. Process models may be updated in part by **feedback** used to observe the controlled process.
- The control structure is hierarchical, meaning that the controllers at the top have a higher authority and control - and as we go down, the level of control and authority decreases.
- All the arrows down are control actions, arrows up are feedback, and arrows in and out of a box (horizontally) are the inputs and outputs. Input and output could signal information and communication that is passed between collaborators. However they are not control actions because they do not pose a constraint on behavior of the controlled process.

At its core, a systems lens for ML applications invites the developer to zoom out from the ML model and situate it in its context of use. The following are guidelines for developing a control structure for ML system. To develop a control structure diagram for a defined system boundary for an ML system:

- Examine and understand how the ML-based system works at a technical and operational level while understanding the context of use and the broad social system. Concretely, gaining this understanding will involve engaging with system developers, users, and a broader set of stakeholders.
- Identify stakeholders that are relevant to the given system boundary.
- Identify all the automated and technical components necessary for the functioning of the given system.
- Outline the responsibilities, key functions, and roles that individuals, groups, and technical component/sub-systems hold for all the elements you identified in the previous two steps. This step also relies on a sufficient understanding of the people, products, and processes involved in the ML development pipeline. Conduct the necessary research to acquire this information.
- Decide if the identified sub-systems (i.e., people, automated system, etc.) are controllers or controlled processes per the definition provided above.
- Determine the hierarchy between various controllers and controlled processes by deciding which sub-system is posing constraints on another sub-system.
- Identify the control actions and feedback (if present) after having the initial hierarchical arrangement.
- Outline the control algorithm and process or mental model for a given controller briefly.

Here is a non-exhaustive list of controllers, control actions, and feedback common to ML systems:

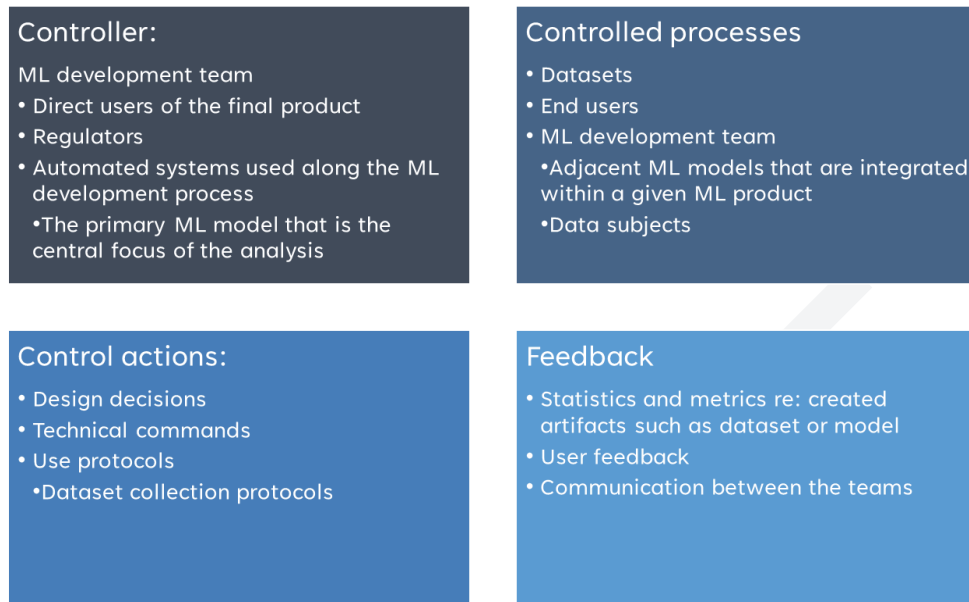
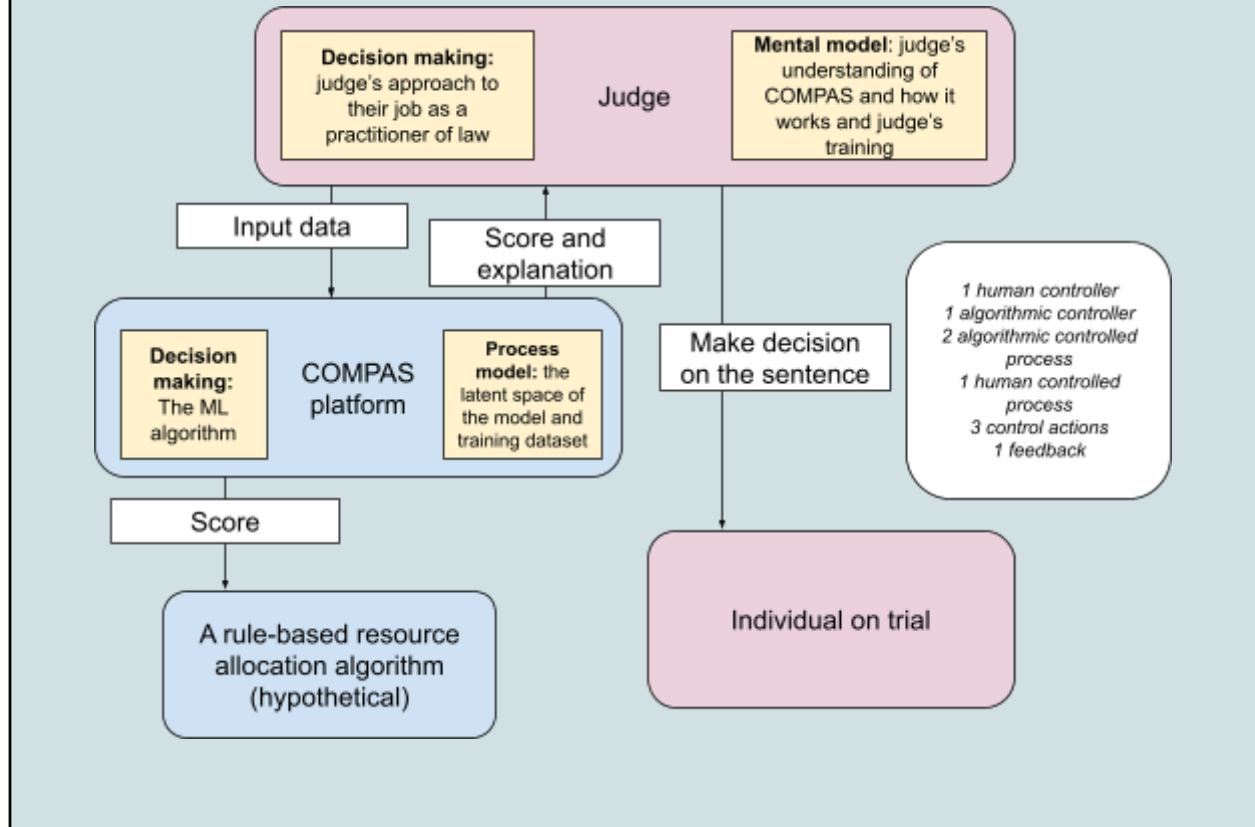


Figure 5. Potential examples of controller, controlled process, control action and feedback for ML systems

The most valuable part of drawing a control structure diagram for a system along the ML development pipeline is that it allows you to get a full perspective on how the different components of an ML pipeline relate to each other. It is well-known that it is impossible/very difficult to develop an abstraction of the complex relationships that exist between decision-makers, teams, automated systems, and end-users in the ML development/deployment process. In the first round of analysis, draw a simple diagram with 3 to 4 controllers and controlled processes that correspond to 4-5 control actions. Control structure diagrams are refined as the system and understanding of that system is developed.

Example:

Simplified control diagram for the use system boundary (S1)



Step 3: Identifying unsafe control actions

Examining unsafe interactions

From the control structure diagram, we use the control actions for step 3. Leveson defines the following key terms for this step:

- **Definition:** An Unsafe Control Action (UCA) is a control action that, in a particular context and worst-case environment, will lead to a hazard. For example, incorrect processing of training data could lead to potential discrimination against certain user groups. In this scenario, incorrect processing covers a broad range of unsafe control actions.
- **Control action can be unsafe in four different ways:**
 - Not providing the control action
 - Providing the control action or providing the incorrect control action
 - Providing a potentially safe control action but too early, too late, or in the wrong order
 - Providing the control action for too long or stopping it too soon (for continuous control actions, not discrete ones).

To develop the list of unsafe control action, compile the list of all the control actions identified in Step 2. For each control action:

- Review the list of hazards and losses from Step 1.
- Review your control structure from Step 2.
- Identify under what conditions could the control action be unsafe given the four different heuristics identified earlier. Think through the following questions:
 - Under what conditions, would **providing** this [CONTROL ACTION] lead to [HAZARD]?
 - Under what conditions, would **not providing or incorrectly providing** [CONTROL ACTION] lead to [HAZARD]?
 - Under what conditions, would **providing** this [CONTROL ACTION] **too early or too late** [IN A GIVEN TIMELINE] lead to [HAZARD]?
 - Under what conditions, would **providing** this [CONTROL ACTION] **too long or stopping it too soon** lead to [HAZARD]?

When thinking about the conditions for potentially unsafe control actions, examine the states of the controllers, controlled process, and controlled actions identified in Step 2. You have completed Step 3 when you have examined all the control actions from Step 2. The output of this step is a list of unsafe control actions that are linked to hazards identified in Step 1 and the control structure developed in Step 2.

A UCA contains five parts: source, type, control action, context, and link to hazard. The first part is the controller that can provide the control action. The second part is the type of unsafe control action (provided, not provided, too early or too late, stopped too soon or applied too long). The third part is

the control action or command itself (from the control structure). The fourth part is the context discussed above, and the last part is the link to hazards (or sub-hazards). The same approach can be applied to human controllers to identify UCAs. The following diagram illustrates what a typical UCA table looks like for a PHASE analysis:

Controller	Controlled process	Control action	Context	Providing	Not providing	Too late or too early	Applied too long or stopped too early	Hazard
Classifier for filtering data	Dataset	Eliminate certain individuals as they are outliers	Recidivism algorithm in criminal justice	The filter eliminated too many data points.	The filter captured too many of the outliers, making it difficult to develop an accurate model.	The filter is applied before the data is cleaned up for missing values.	The filter keeps eliminating the data from continuous use of the system.	A low-quality training dataset is generated.

Notably, the ML development process - especially the model development and data processing is highly iterative. The developer of the system is generally rapidly iterating on the model architecture, choices of hyperparameters, etc. Furthermore, ML systems themselves are probabilistic in nature, and the control action that might be given from a given system could change over time. Furthermore, some generative systems develop emerging capabilities that could present themselves as emerging control actions. Therefore, to think through potential unsafe control actions, it is important to consider the unique properties of ML systems/development processes by carefully thinking through the *type* and *context* of the unsafe control action.

Example:

Controller	Controlled process	Control action	Context	Providing	Not providing	Too late or too early	Applied too long or stopped too early	Hazard
Judge	COMPAS platform	Input data	The judge inputs the required data into the COMPAS platform.	The judge enters incorrect information	The judge does not provide the key information	Not applicable	Not applicable	H3. The judge misuses the system.

Step 4: Identifying loss scenarios

Identifying potential causal scenarios

Once the UCAs are identified, in the last step, you will brainstorm and identify all the potential causal scenarios for each UCA. These are called loss scenarios - the circumstances that could potentially lead to a UCA, which could ultimately lead to a loss. To identify the loss scenarios:

- Examine all the components of a control structure diagram, including the exchange between different sub-systems and the inner workings of each system.
- Examine controller behavior by studying potential issues related to the controller by identifying 1) inadequate/incorrect controller algorithm or decision-making process, 2) inadequate/incorrect process or mental model.
- Examine potential avenues of inadequate feedback, which can mean that either feedback is not received or sent. Alternatively, the received feedback is not adequate or correct.

Each UCA generally corresponds to multiple loss scenarios. Typically it is relatively easy to identify at least 2 to 5 loss scenarios per UCA.

Example:

Unsafe control action	Loss Scenarios
The judge enters incorrect information	The judge's mental model of the COMPAS platform is incorrect and they do not know what data is needed for the COMPAS platform to work.
	The system does not provide any feedback about the quality of the data inputted by the judge. It just provides the predicted score.
The judge does not provide the key information.	The COMPAS platform does not have the means of inputting of all the data.
	The platform does not provide any feedback about the quality of the data inputted by the judge. It just provides the predicted score.

Now what: how do we synthesize the findings from PHASE?

After completing the first iteration of these guidelines on a given ML application, you will have the following:

- A list of system-level losses and hazards that are linked to each other
- A control structure diagram showing the key controllers, controlled processes, control actions and feedback loops
- A list of unsafe control actions
- A list of loss scenarios for each unsafe control action.

The next step is to synthesize and prioritize this information in the form of safety requirements, which outline the necessary constraints for ensuring that a given system is safe (i.e., will not lead to the losses identified for the given system). Development of safety requirements starts by closely examining the loss scenarios and seeing what types of constraints could minimize or eliminate the possibility of unsafe control actions in those scenarios. Sometimes, safety requirements are also oriented toward minimizing harm from an unsafe control action if it does occur. For ML systems, these requirements can be technical in nature (i.e., using or not using specific features in a dataset or using ML models optimized for fairness metrics), institutional (i.e. establishing specific product management and communication protocols between model and product development teams), or regulatory (i.e., requiring ML-based products to go through conformity assessments). Developing requirements is beyond the scope of this guideline, and it will not be a part of the assignment accompanying this guideline.

Case study: Predictive Algorithm in Clinical Decision Making

Overview of the case study

The use of predictive algorithms in clinical diagnosis is a common application of machine learning algorithms. In this case study, we perform STPA on an ML algorithm used to predict the likelihood of late-onset sepsis (LOS) in pre-term infants before a clinical suspicion. This algorithm could be used as an early warning system for clinicians to start early diagnostic testing for LOS and, ultimately, a quicker treatment. Clinicians could use this algorithm as a tool for “earlier than suspected detection of LOS in a group of very preterm infants.” This algorithm is used in a human-in-the-loop setting where a clinician will always need to make a decision based on the output of the algorithm. In this case study, the predictive algorithm is being developed at a hospital based in the Netherlands and is currently running in a shadow mode for research purposes where the clinicians are not actively using the system as part of their diagnosis.

Neonatal sepsis is a highly infectious illness that poses a major threat to the health and survival of newborns on a global scale. In particular, premature infants and infants with very low birth weight (VLBW) are susceptible to neonatal sepsis. Neonatal sepsis is typically classified based on when it occurs. Early onset sepsis (EOS) develops in the first few days of life and is primarily caused by “bacteria transmitted from the mother to the baby during labor or the prenatal period”. Late-onset sepsis (LOS) occurs after the first week of life and can be attributed to bacteria acquired from the baby's environment. The ramifications of late-onset sepsis (LOS) encompass neurological impairment, bronchopulmonary dysplasia, extended hospitalization, multi-organ system failure, and mortality. On a global scale, neonatal sepsis ranks as the third leading cause of neonatal deaths, accounting for 13% of them. Despite proactive treatments in high-income countries, 39% of all neonatal sepsis cases result in significant disability or death.

The timely diagnosis of LOS is difficult because its symptoms are quite comparable to other types of neonatal conditions. Furthermore, laboratory tests (ex. Positive blood culture) provide limited value as they illustrate the disease when it has progressed quite significantly. Considering the importance of early diagnosis and detection, clinicians often prescribe a range of antibiotics as they suspect an LOS diagnosis based on clinical signs. However, early prescription of antibiotics has its own negative side effects on infants. Most recently, clinicians have been looking at diagnosing LOS at an earlier time via other methods, including predictive ML algorithms.

Tags: [#supervised learning](#), [#human-in-the-loop](#), [#research](#), [#Europe](#)

Approach to analysis

For this application of STPA, the lead analyst gathered the necessary information from published articles and conversations with two experts who were involved in building this ML system. Two advisors also provided feedback on the drafts of the STPA analysis and this feedback was incorporated for the final analysis. The three main scopes that were examined for this analysis include **1) data collection and processing, 2) model development and evaluation and 3) clinical implementation.** The focus of the data collection and processing was to develop the appropriate training dataset for model development that has a positive clinical impact. The model development and evaluation stage considered key factors such as model interpretability, including features that detect actionable insights about the patient's well-being and creating model outputs that have potential use in clinical practice. Even though the clinical implementation of this project is not fully developed, in this analysis, we use the simulated clinical use as the baseline for our analysis.

Summary of findings

Scope 1 - Data collection and processing

Step 1: Identify purpose of analysis

Losses:

The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology).

Stakeholders: patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team

Values: patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency,

ID Loss

- L1 Loss of life/injury
- L2 Loss of time
- L3 Loss of reputation/credibility
- L4 Loss of patient privacy
- L5 Loss of quality

System boundary: The focus of the data collection and processing was to develop the appropriate training dataset for model development that has a positive clinical impact.

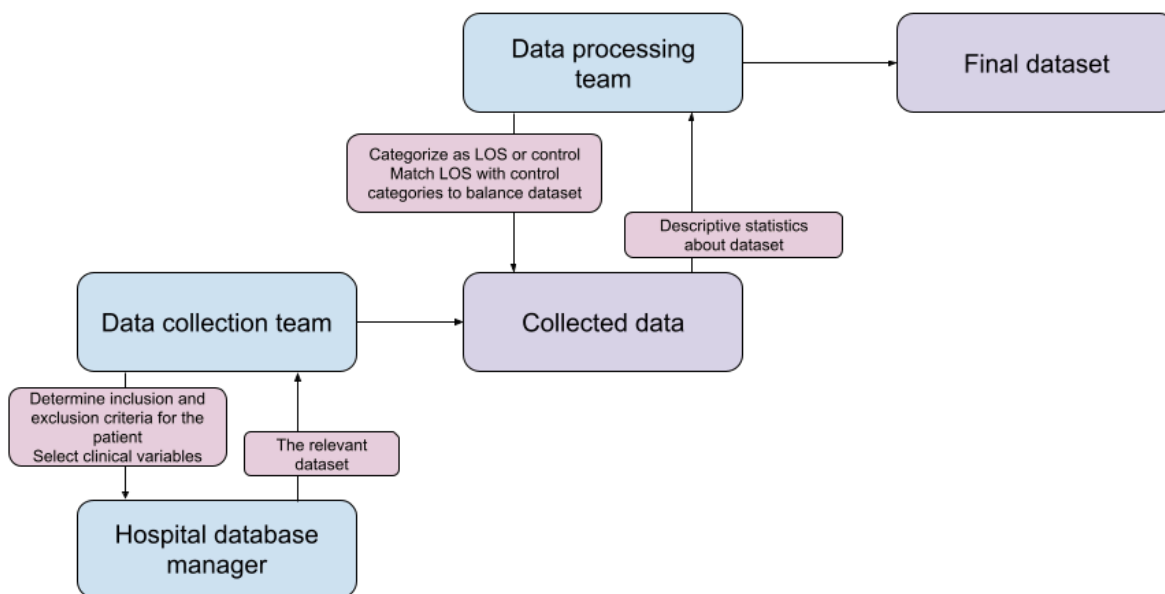
System- level hazards:

- H1. The system creates a training dataset that does obide by the relevant data privacy regulations.
- H2. The system creates a training dataset that is unbalanced across relevant labels.
- H3. The system creates a training datasets that is mislabeled.
- H4. The system creates a training dataset training set contains features from clinical data that are exist due to physicians suspicion of LOS.
- H5. The system creates at training dataset that is too small for training a model training

ID	Hazards	Associated Losses				
		L1	L2	L3	L4	L5
H1	The system creates a training dataset that does obide by the relevant data privacy regulations	FALSE	FALSE	TRUE	TRUE	TRUE
H2	The system creates a training dataset that is unbalanced across relevant labels.	TRUE	TRUE	TRUE	FALSE	TRUE
H3	The system creates a training datasets that is mislabeled.	TRUE	TRUE	TRUE	FALSE	TRUE
H4	The system creates a training dataset training set contains features from clinical data that are exist due to physicians suspicion of LOS.	TRUE	FALSE	TRUE	FALSE	TRUE
H5	The system creates at training dataset that is too small for model training .	FALSE	TRUE	FALSE	FALSE	TRUE

Step 2: Create control structure diagram

The control structure diagram consists of three human controllers and 2 artifacts that are controlled and output/inputs of other controllers. Based on the information gathering, the data collection and processing scope did not include any automated processes.



The three human controllers include 1) the data collection team, 2) the hospital database manager, and 3) the data processing team. This scope does not include other relevant actors, such as clinicians or the individuals who are in charge of ensuring that data privacy protocols are followed within a hospital research team. We focus on analyzing four control actions for the next step, which include 2 control actions between the data collection team and hospital database manager and 2 others between the data processing team and collected data. It is noteworthy that the data processing and collection team consisted of the same people for this project.

Step 3: Identify unsafe control actions

We brainstormed a preliminary list of nine unsafe control actions following the original STPA heuristics. The UCAs lead to at least one hazard. The unsafe control actions were focused on the different mistakes and issues that could happen when selecting the relevant data from the hospital database and processing it to prepare the training/evaluation dataset. These include but are not limited to 1) not following the inclusion criteria or having inappropriate inclusion criteria, 2) selecting improper clinical variables to be included as features in the collected dataset, 3) miscategorizing the collected dataset as LOS or control, and 4) balancing the dataset improperly.

Step 4: Identify loss scenarios

We uncovered four overarching loss scenarios, including 1) Data collection/processing team did not engage with the necessary stakeholders (i.e. physicians) to understand what type of data needs to be collected or how the data needs to be processed, 2) Improper inspection/feedback/descriptive statistics about the data being collected or processed, 3) Inadequate data anonymization practice in the hospital, and 4) Below the standard approaches to data processing (i.e. poor data balancing)

Scope 2 - Model development and evaluation

Step 1: Identify purpose of analysis

Losses:

The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology).

Stakeholders: patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team

Values: patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency,

ID	Loss
L1	Loss of life/injury
L2	Loss of time
L3	Loss of reputation/credibility
L4	Loss of patient privacy
L5	Loss of quality

System boundary: The model development and evaluation stage considered key factors such as model interpretability, including features that detect actionable insights about the patient's well-being and creating model outputs that have potential use in clinical practice

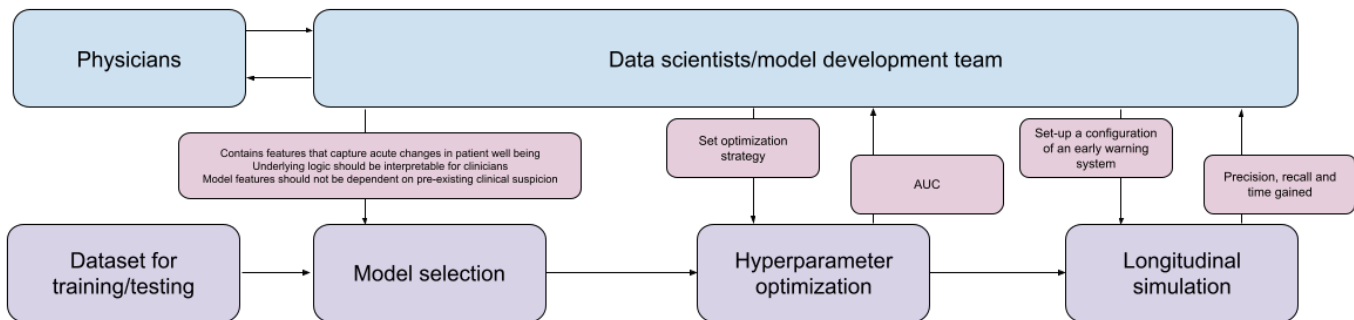
System- level hazards:

- H1. The system creates a model/output that does not meet performance requirements (ex. accuracy, precision and recall).
- H2. The system creates a model/output that is not interpretable.
- H3. The system does not evaluate adequately/correctly for performance of the model and output.
- H4. The system does not evaluate adequately/correctly for interpretability the model and output.
- H5. The system does not evaluate adequately/correctly for privacy and fairness the model and output.

ID	Hazards	Associated Losses				
		L1	L2	L3	L4	L5
H1	The system creates a model/output that does not meet performance requirements (ex. accuracy, precision and recall).	TRUE	TRUE	TRUE	FALSE	TRUE
H2	The system creates a model/output that is not interpretable.	TRUE	TRUE	TRUE	FALSE	TRUE
H3	The system does not evaluate adequately/correctly for performance of the model and output.	TRUE	FALSE	TRUE	FALSE	TRUE
H4	The system does not evaluate adequately/correctly for interpretability the model and output.	TRUE	FALSE	TRUE	FALSE	TRUE
H5	The system does not evaluate adequately/correctly for privacy and fairness the model and output.	TRUE	FALSE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

The control structure consists of 2 human sub-systems and 4 technological artifacts/processes. The main controller here is the data scientist or the full model development team. This controller collaborates with the physician as indicated by the input/output arrows and has control over three main process/artifacts of model selection hyperparameter optimization and longitudinal simulation. Dataset for training and testing is an input to the model selection process. The output of all the purple boxes is an input for the next one.



Step 3: Identify unsafe control actions

The STPA identified twelve Unsafe Control Actions (UCAs) within the model development and evaluation process, focused on three key areas: model selection, hyperparameter optimization, and longitudinal simulation. Unsafe actions included failing to provide clinically meaningful features, relying on biased or implausible data, applying misaligned optimization strategies, and simulating model behavior under unrealistic assumptions. Many UCAs stemmed from poor timing (e.g., decisions made without clinician input), incorrect content (e.g., label leakage), or insufficient oversight in how configurations were defined and tested. These control flaws often reflect breakdowns in feedback loops, incomplete mental models of the clinical environment, or misaligned priorities between model performance and real-world usability.

Step 4: Identify loss scenarios

The loss scenarios observed in the model development and evaluation scope reveal how system-level failures can emerge from control structure breakdowns. Many losses stemmed from missing feedback loops—for example, when model developers made architectural or optimization decisions without timely clinician input, leading to models that were uninterpretable or clinically misaligned. In other cases, loss resulted from incorrect mental models held by developers, such as overestimating model generalizability or misunderstanding how features function in practice. This led to the inclusion of misleading or biased inputs and evaluation strategies that failed to reflect real clinical needs.

Flawed process models also played a role: simulations configured with idealized data or assumed clinician behavior failed to account for the complexity and variability of real-world care. This misrepresentation affected how model outputs were interpreted and trusted, degrading quality (L5) and credibility (L3). Additionally, poorly designed control algorithms, such as hyperparameter optimization routines that prioritized aggregate accuracy over clinically meaningful trade-offs, resulted in models that underperformed in early detection or treated different patient subgroups unfairly—exacerbating risks to life (L1), time (L2), and equity (L5).

Overall, these scenarios demonstrate that technical choices made during model development are deeply shaped by system control dynamics. Without accurate process models, bidirectional feedback, and control logic that reflects clinical goals, even high-performing models can lead to serious downstream harm.

Scope 3 - Clinical implementation

Step 1: Identify purpose of analysis

Losses:

The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology).

Stakeholders: patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team

Values: patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency,

ID Loss

- L1 Loss of life/injury
- L2 Loss of time
- L3 Loss of reputation/credibility
- L4 Loss of patient privacy
- L5 Loss of quality

System boundary: Use the predictive algorithm to deliver quality patient care and treat or order necessary tests as early as possible.

System- level hazards:

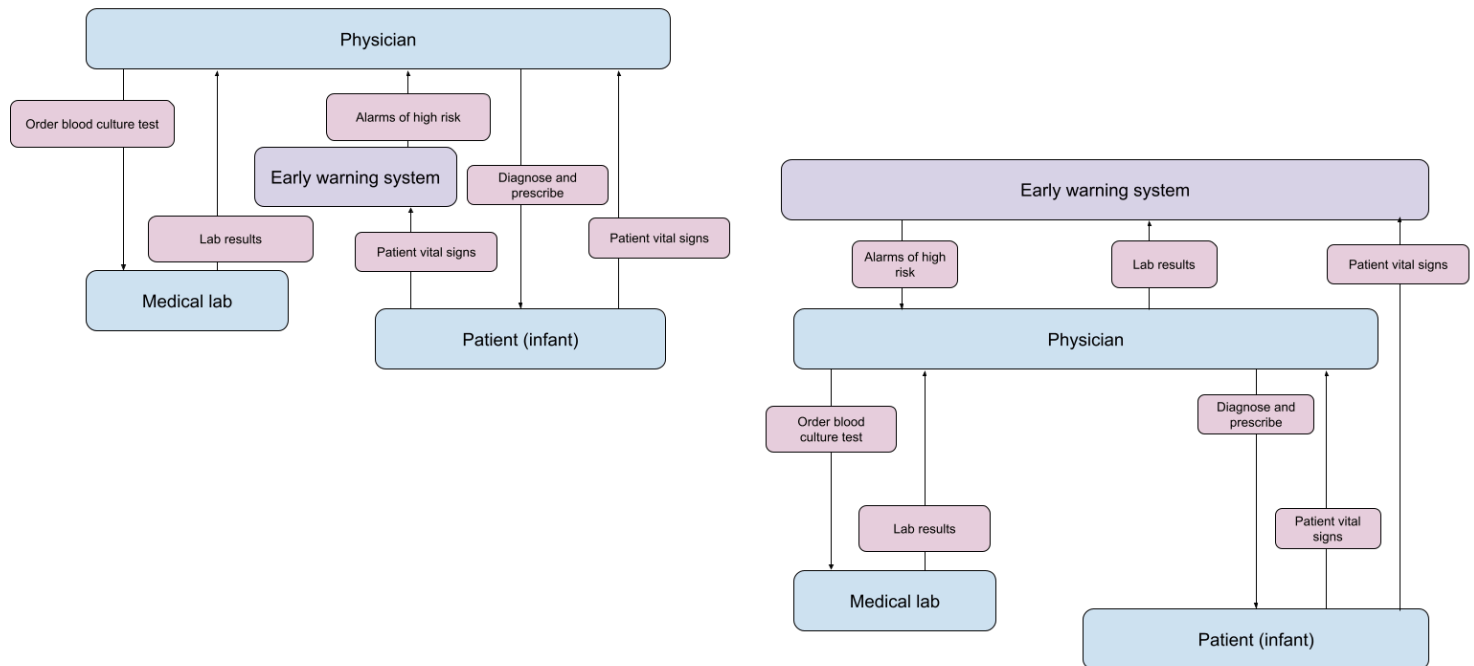
- H1. The physician does not understand or misunderstands the output of the predictive algorithm.
- H2. The physician misdiagnosis for LOS and prescribes antibiotics.
- H3. The physician diagnosis LOS too late.

ID	Hazards	Associated Losses				
		L1	L2	L3	L4	L5
H1	The physician does not understand or misunderstands the output of the predictive algorithm.	TRUE	TRUE	TRUE	FALSE	TRUE
H2	The physician misdiagnosis for LOS and prescribes antibiotics.	TRUE	FALSE	TRUE	FALSE	TRUE
H3	The physician diagnosis LOS too late.	TRUE	FALSE	TRUE	FALSE	TRUE

Step 2: Create control structure diagram

The control diagram has three human/group sub-systems and one automated system. Depending on whether the physicians have to use the alarm system or not the set-up of the control hierarchy could be different. In the left control diagram the physician is the highest level of controller because they do not need to use the early warning system. They just get feedback

from it in form of alarms. Whereas, in the right scenario the physicians are required to use the early warning system and there the system is higher in the hierarchical control structure. The control actions are different in each case. The case to the left has 2 main control actions as opposed to three control actions for the control diagram to the right. The nurse and the parent/caretaker will likely be involved in this scope but for now we have kept them outside for simplicity of the analysis.



Step 3: Identify unsafe control actions

In our STPA of a predictive early warning system for Late-Onset Sepsis (LOS) in preterm infants, we identified twelve Unsafe Control Actions (UCAs) across three key control relationships:

1. Physician → Medical Lab (Order Blood Culture Test):

UCAs occur when the physician fails to order a test when needed, orders it unnecessarily, delays it, or orders it prematurely. These unsafe actions can stem from misinterpretation of risk signals or overreliance on the algorithm, leading to missed diagnoses or over-testing.

2. Physician → Patient (Diagnose and Prescribe):

Unsafe actions include failing to diagnose or prescribe treatment when LOS is present, doing so prematurely or based on misinterpretation, or delaying action. These behaviors are particularly risky given the narrow treatment window for LOS.

3. Early Warning System → Physician (Alarm of High Risk):

UCAs arise when the system fails to generate an alarm when risk is high, raises a false alarm, delays the alarm, or issues too many alerts. These failures can result in missed interventions or alarm fatigue, undermining clinician trust

and response behavior.

Step 4: Identify loss scenarios

The loss scenarios identified in this STPA reflect systemic vulnerabilities that can be traced to control flaws in feedback, process models, and control algorithms — all of which shape the behavior of both human and automated components in the system.

- **Missed or delayed detection of LOS** (e.g., the physician fails to act or the system fails to alarm) often stems from missing or delayed feedback loops between the early warning system and the physician. If the alarm is not triggered or is delayed, the physician's process model may not reflect the actual physiological state of the infant, leading to underreaction and loss of life or health (L1).
- **Unnecessary testing or treatment** (e.g., false positives or premature test orders) frequently result from an **incorrect control algorithm** in the early warning system or a mismatched mental model on the part of the physician. If the algorithm over-predicts risk or the physician does not correctly interpret the model's output, it can lead to inefficient use of resources (L2), harm from overtreatment (L1), and reduced trust in both the physician and the system (L3, L5).
- **Delayed diagnosis or prescription** may be caused by a misalignment between the physician's process model and the true system state. If the physician assumes low risk due to a lack of alerts or unclear decision thresholds, their timing of intervention may not match the needs of the patient, resulting in harm and reputational loss.
- **Alarm fatigue and clinician desensitization** arise when the control algorithm lacks contextual awareness or adaptive thresholds, leading to frequent false or low-value alarms. This breakdown in the control loop between the system and the user causes clinicians to ignore or disable alerts, effectively breaking the feedback channel and increasing the likelihood of missed true positives (L1, L3, L5).
- **Breakdowns in clinician-system coordination** often reflect incomplete or ambiguous system feedback, where alerts or outputs lack actionable guidance, or are not well-integrated into clinical workflow. These issues suggest a need for better designed process models within both the system and the training clinicians receive on how to use it.

Case study: RL use in Insulin Injection

Overview of the case study

In this application of STPA, we are investigating potential hazards from using an automated insulin injection system that uses a reinforcement learning algorithm. Insulin injection is a critical part of treating patients with diabetes. Currently, the appropriate dosage of insulin and other drugs is studied in clinical trials. It is well-reported that considering the inherent limitations/requirements in clinical trials, only a limited number of people can participate in clinical trials. Often, people with comorbidities and from minority populations do not meet participation requirements. The appropriate dosage determined from the clinical trials is generally optimal for people who are most similar to the group that got to participate in the trials. This poses a health risk to people who are not part of this group. Considering the limitations posed by clinical trials and other adjacent issues the medical community is exploring the use of RL algorithms for personalization of drug dosage as one type of a solution.

Currently, insulin injection is currently being done manually, where the patient is responsible for monitoring their glucose level after meals and injecting on a needed basis. Alternatively, patients with the appropriate socioeconomic means use an artificial pancreas, which facilitates insulin injection automatically. Currently, the algorithmic system used in artificial pancreas does not use machine learning algorithms. The motivation behind using RL algorithms in insulin injection is to personalize, optimize, and fine-tune the amount of insulin dose that is needed for each individual person based on their specific reaction (i.e. change in glucose level) to what they eat and their environmental conditions. To illustrate what we could learn from an application of STPA for such a system, we consider three different scopes of analysis, including **1) Implementation of RL algorithm in an artificial pancreas, 2) Design of RL algorithm, 3) Use of RL-based artificial pancreas in patient care.**

Tags: [#reinforcement learning](#), [#human-out-of-the-loop](#), [#research](#), [#Canada](#)

Approach to analysis

The analysis was led by the lead author, who collaborated with two experts in reinforcement learning. One of the analysts focused on developing RL algorithms for insulin injection and had worked extensively with physicians in the field to understand the clinical setting and impact of this technology on potential users. The three analysts met on a bi-weekly basis from February 2023 to Aug 2023 and worked on two different STPA analyses following the steps outlined in the STPA handbook ([Leveson 2016](#)), which was supplemented by the lead author's expertise in applying the STPA process. It is important to note that use of RL in insulin injection is at the research stage, and the analysis did not

consider operational settings for implementing such a technology. The focus of this analysis was to examine how the STPA process applies to ML systems that are based on RL algorithms.

Summary of findings

Scope 1 - Implementation of RL algorithm in an artificial pancreas

Step 1: Identify purpose of analysis

Losses:

From the stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money.

Stakeholders: We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values

Values: Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income.

ID	Loss
L1	Loss of life/injury
L2	Loss of quality of life (efficiency, autonomy, amount of effort)
L3	Loss of efficiency
L4	Loss of money

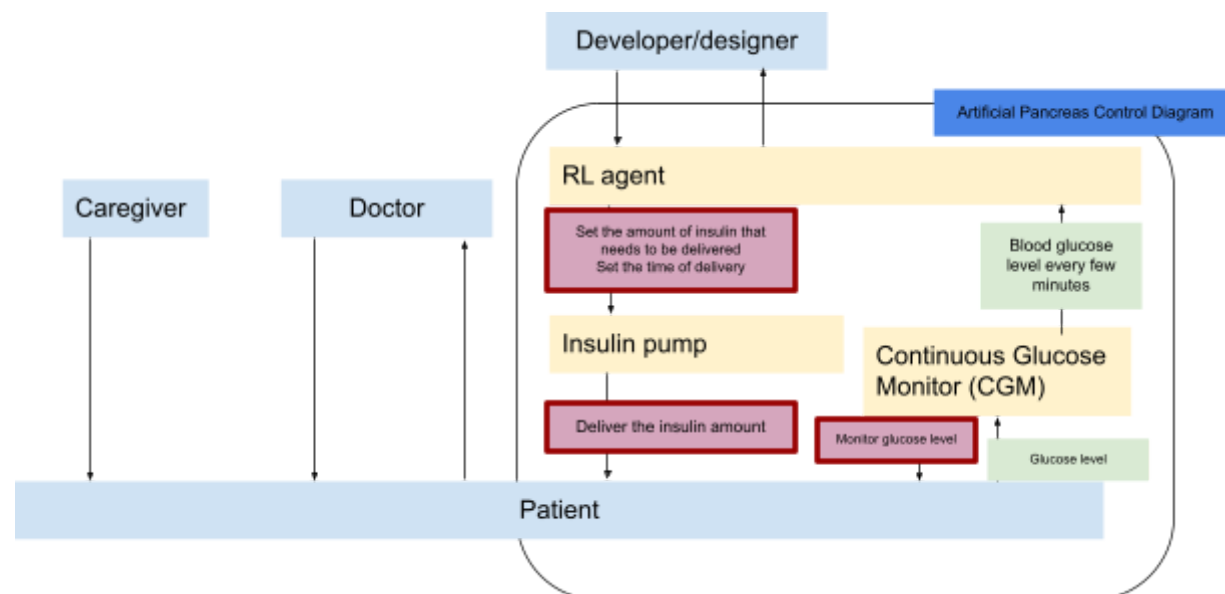
System boundary: For the given scope, the goal of the system is to deliver the right dosage of insulin at the needed time for all patients.

System- level hazards:

- 1) Artificial pancreas provides a larger dose than needed at a given time to the patient.
- 2) Artificial pancreas provides a smaller dose than needed at a given time to the patient .

ID	Hazards	Associated Losses			
		L1	L2	L3	L4
H1	Artificial pancreas provides a larger dose than needed at a given time to the patient.	TRUE	TRUE	TRUE	TRUE
H2	Artificial pancreas provides a smaller dose than needed at a given time to the patient .	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram



When modelling the control structure diagram we identified 3 automated controllers (RL agent, insulin pump and continuous glucose monitor) and one human controller (patient). The patient is not actively taking part of controlling any part of this process – they are the receiver of decisions made by the RL agent and provide of data through the CGM. There are four main control actions that we examined in this scope including 1) RL agent to pump: set the amount of insulin that needs to be delivered, 2) RL agent to pump: set the time of delivery and 3) Pump to patient: deliver the insulin amount 4) CGM to patient Monitor glucose level. There is no direct feedback loop from the patient to the pump or the RL agent. The CGM scans the patient's glucose level and sends that to the RL agent.

Step 3: Identify unsafe control actions

The STPA analysis identified 17 Unsafe Control Actions (UCAs), primarily focused on functional performance hazards: either too much or too little insulin being delivered. The key UCAs reflect breakdowns in coordination among automated components (RL agent, insulin pump, CGM) and lack of human-in-the-loop oversight. The four critical control actions considered were:

1. RL Agent → Pump: Set insulin amount
2. RL Agent → Pump: Set delivery time
3. Pump → Patient: Deliver insulin
4. CGM → RL Agent: Monitor glucose level

Examples of UCAs include:

- RL agent sets too large a dose or delays insulin delivery, causing hypo- or hyperglycemia (linked to Hazard 1 or 2).
- The pump delivers insulin at the wrong time or fails to deliver, due to incorrect or missing commands.
- CGM provides inaccurate glucose data, which misguides the RL agent's decision-making.
Importantly, no UCAs in this scope directly related to social or ethical risks were identified. Instead, the risks were tied to automated functional failure, stemming from missing or incorrect control actions, inadequate timing, or inaccurate sensor data.

Step 4: Identify loss scenarios

After examining potential controller behaviours and existing feedback loops, we brainstormed an initial list of 7 loss scenarios that apply across numerous UCAs. The losses include 1) Improper design and training of RL algorithm, 2) Incorrect observations due to a faulty CGM, 3) Improper design of the reward function, 4) Pump, valves or tubes in the system are physically damaged and not functioning properly, 5) The pump controller is out of tune with the input from the RL agent., 6) CGM is physically broken, 7) The CGM controller loop is sampling at a higher or a lower frequency than expected. These loss scenarios draw attention to potential design and manufacturing issues of the various controllers.

Scope 2 - Design of RL model

Step 1: Identify purpose of analysis

Losses:

From the aforementioned stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money.

Stakeholders: We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values

Values: Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income.

ID Loss

- L1 Loss of life/injury
- L2 Loss of quality of life (efficiency, autonomy, amount of effort)
- L3 Loss of efficiency
- L4 Loss of money

System boundary: Design and train an RL model which performs well with simulated clinical data and environment.

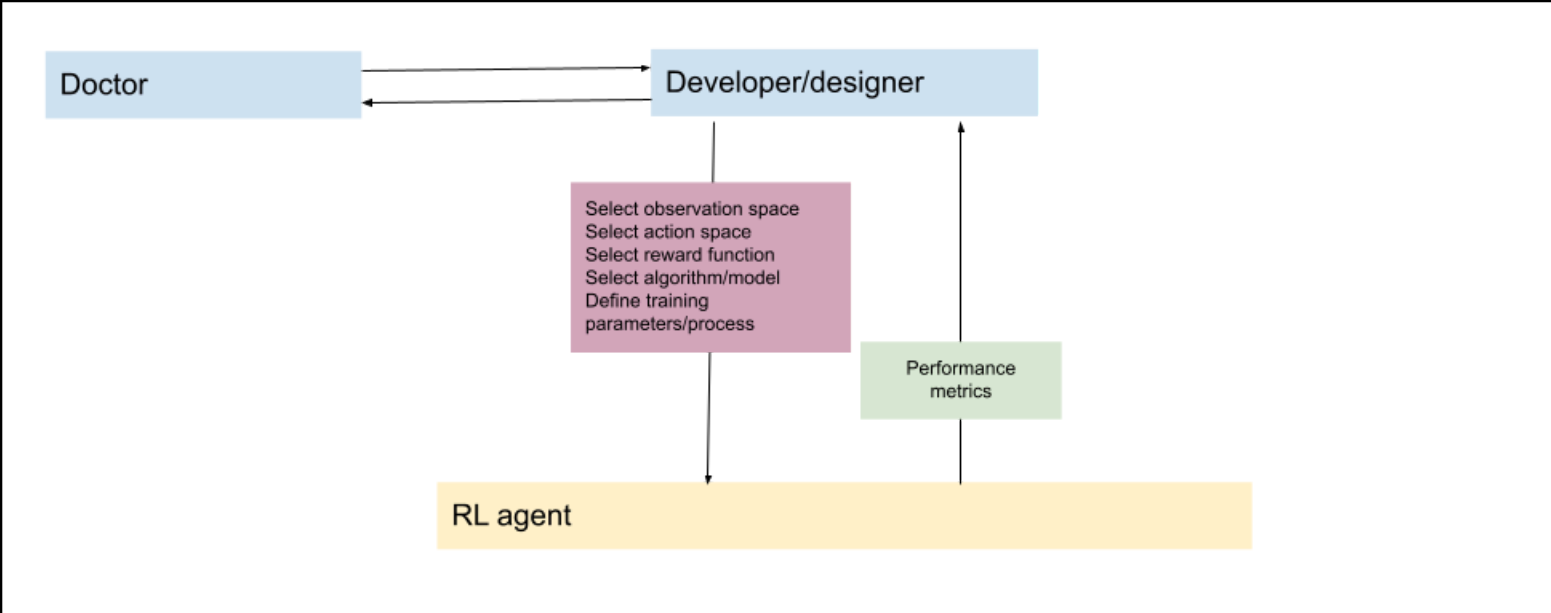
System- level hazards:

1. RL model provides a larger dose than needed at a given time (in simulation environment) .
2. RL model provides a smaller dose than needed at a given time (in simulation environment) .

ID	Hazards	Associated Losses			
		L1	L2	L3	L4
H1	RL model provides a larger dose than needed at a	TRUE	TRUE	TRUE	TRUE
H2	RL model provides a smaller dose than needed at a	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

We have kept this diagram relatively simple. The main control is in the developer/designer of the system where they make the design decisions for the RL agent. There are no other ML models that are being used to make this decision for the system. The doctor provides feedback to the developer based on their expertise in their collaboration.



Step 3: Identify unsafe control actions

In Scope 2, Unsafe Control Actions center around the design decisions made by the developer of the RL model used in an artificial pancreas. These UCAs occur during the specification of the reward function, construction of the training environment, selection of input features (state representation), tuning of hyperparameters, and decisions about when and how to evaluate the model. Each control action, when performed incorrectly or omitted, can lead to either over- or underestimation of insulin needs in simulated environments, which poses downstream risks once deployed. All UCAs were tied to two functional hazards: the RL model recommending a larger or smaller dose than required. Notably, UCAs here reflect design-stage control flaws that are less visible in real-time operation but crucially shape system safety.

Step 4: Identify loss scenarios

The identified loss scenarios in Scope 2 emphasize how design flaws in the RL model can propagate into clinical harm. For example, an improperly specified reward function may prioritize short-term glucose drops at the expense of long-term patient health, while a narrow or biased training environment may exclude data from underrepresented populations, leading to ineffective dosing policies. Other scenarios include premature model evaluation, unstable convergence due to poorly tuned exploration parameters, or exclusion of key physiological indicators in the input state. Many of these scenarios stem from missing clinical feedback, biased assumptions about patient diversity, and insufficient validation procedures. Although these issues arise in simulation, they carry forward to deployment—making design-time decisions a critical site for hazard mitigation.

Scope 3 - Use of RL-based artificial pancreas in patient care.

Step 1: Identify purpose of analysis

Losses:

From the aforementioned stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money.

Stakeholders: We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values

Values: Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income.

ID Loss

- L1 Loss of life/injury
- L2 Loss of quality of life (efficiency, autonomy, amount of effort)
- L3 Loss of efficiency
- L4 Loss of money

System boundary: The physician need to prescribe and manage/monitor a proper diabetes management plan for the patient. The family also needs to help with the management.

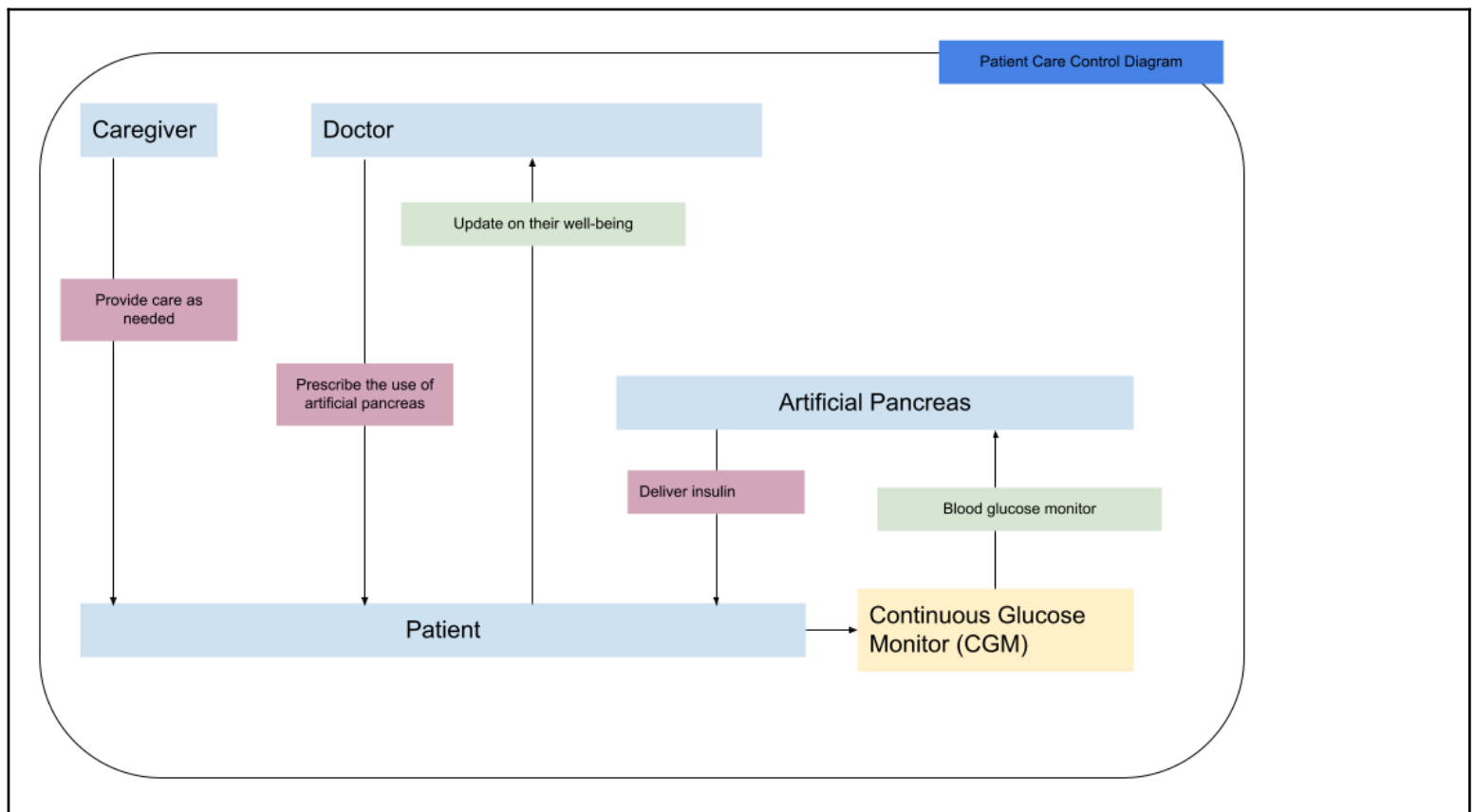
System- level hazards:

- 3. Physician misdiagnoses the patient's condition.
- 4. Physician cannot monitor the patient's status.

ID	Hazards	Associated Losses			
		L1	L2	L3	L4
H1	Physician misdiagnoses the patient's condition.	TRUE	TRUE	TRUE	TRUE
H2	Physician cannot monitor the patient's status.	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

There are four main components - three sub-systems are humans and one is the artificial pancreas. All are higher in the control hierarchy compared to the patient. The care giver is likely lower in control hierarchy than what is actually illustrated here.



Step 3: Identify unsafe control actions

In Scope 3, the UCAs focus on the interaction between human and automated agents in real-world deployment. Three key control actions were analyzed: (1) caregivers providing care to patients, (2) physicians prescribing the artificial pancreas (AP), and (3) the AP delivering insulin. Unsafe control actions included caregivers failing to intervene or acting incorrectly, physicians prescribing the device to unsuitable patients or avoiding its use when beneficial, and the AP delivering insulin inappropriately or not at all. These UCAs highlight failures not only in functional performance but also in trust, training, and the social infrastructure surrounding device use. Both system-level hazards—misdiagnosis and lack of monitoring—were triggered by these unsafe control actions, particularly when feedback between components was missing or ineffective.

Step 4: Identify loss scenarios

The loss scenarios in Scope 3 reflect how poor coordination, device failures, and human misunderstandings can lead to serious health outcomes. For example, caregivers may misinterpret device alerts or overlook malfunctions, leading to incorrect or missing interventions. Physicians may over-rely on or avoid recommending the AP, causing mismatches between patient needs and technology capabilities. On the technical side, the RL-based artificial pancreas may misdeliver insulin due to incorrect input interpretation or physical failures like sensor detachment or battery depletion. These scenarios reveal how fragile system safety becomes when multiple human and technical actors interact without robust safeguards or shared understanding. While the RL model might perform well in simulation, these real-world loss scenarios emphasize the need for user training, system monitoring, and contextual awareness to ensure safe and equitable care.

Case study: T2I use in creative practice

Overview of the case study

Applying STPA at three stages of the ML development pipelines enabled analysts to uncover a wide range of failures and hazards without the need for detailed information on the specific T2I model. In particular, the three scopes of analysis (training data processing, product integration, and end use) each allowed practitioners to look beyond a specific model in isolation and focus on processes and interactions as it is integrated with other actors and technical systems. In the next three sections, we reflect on the failures and hazards discovered at each stage. The analysis uncovered known failures and hazards (e.g., the creation of non-consensual sexual imagery) and novel ones that have not been recognized to the best of our knowledge (e.g., the impact of English-word filters on lexical change). We also identified failures and hazards that could present social and ethical risks for different actors, which can inform the prioritization of mitigation strategies.

Approach to analysis

STPA could be applied to various scopes of analysis that require a different set of information. Scoping an ML system for such an analysis is a non-trivial task. An ML system can be divided into its component parts (e.g., training data, model, user interface), development process (e.g., training, testing, early deployment), stakeholders involved (e.g. ML developers, a community of developers interfacing and building on the model APIs, end-users), and so on. The results of the analysis can be drastically different depending on the chosen scope of analysis [10, 42].

Based on the information gathered from the interviews with T2I experts and workshops with artists, we identified three scopes of analysis along the ML development pipeline:

- Scope 1: the data processing necessary for creating a training dataset
- Scope 2: how a T2I model is integrated into a production environment along with other ML models
- Scope 3: how an artist uses T2I model demo as part of their creative process

Selecting the scopes along the ML development pipeline allows for the examination of critical processes and interactions as discussed by previous scholarship [31, 67, 70]. We chose to focus on these scopes of analysis because we had access to the most amount of publicly available and shareable information about the elements involved.

The STPA analysis was led by the first author of this paper and three of the co-authors provided feedback on iterations of the analysis. A separate analysis was conducted for each one of the scopes. In total three STPA analyses were conducted. The lead author spent somewhere between 10 - 12 hours implementing the STPA process on one scope. All sources of data were used as input for STPA process. The STPA analysis resulted in a list of unsafe control actions and corresponding safety requirements.

We followed the original STPA process for each one of the scopes (as described below) and did not alter them to specifically uncover social and ethical risks.

Summary of findings

Scope 1 - data processing

Step 1: Identify purpose of analysis

Losses:

From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.**

Stakeholders: From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68]

Values: Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist's reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company's reputation.

ID Loss

- | | |
|----|---|
| L1 | Loss of creativity |
| L2 | Loss of diversity in generated image |
| L3 | Loss of accessibility (who can use these systems) |
| L4 | Loss of efficiency (in generating image) |
| L5 | Loss of quality |
| L6 | Loss of reputation |

System boundary: In our application, we bound the data processing stage to start from the identification of one or multiple source datasets and end with the creation of a training dataset ready for model development purposes. In this scope, we do not directly analyze the steps involved in how the data sources were obtained and who was involved in the data collection process. The focus is on how the data sources are processed for creating a training dataset.

System-level hazards:

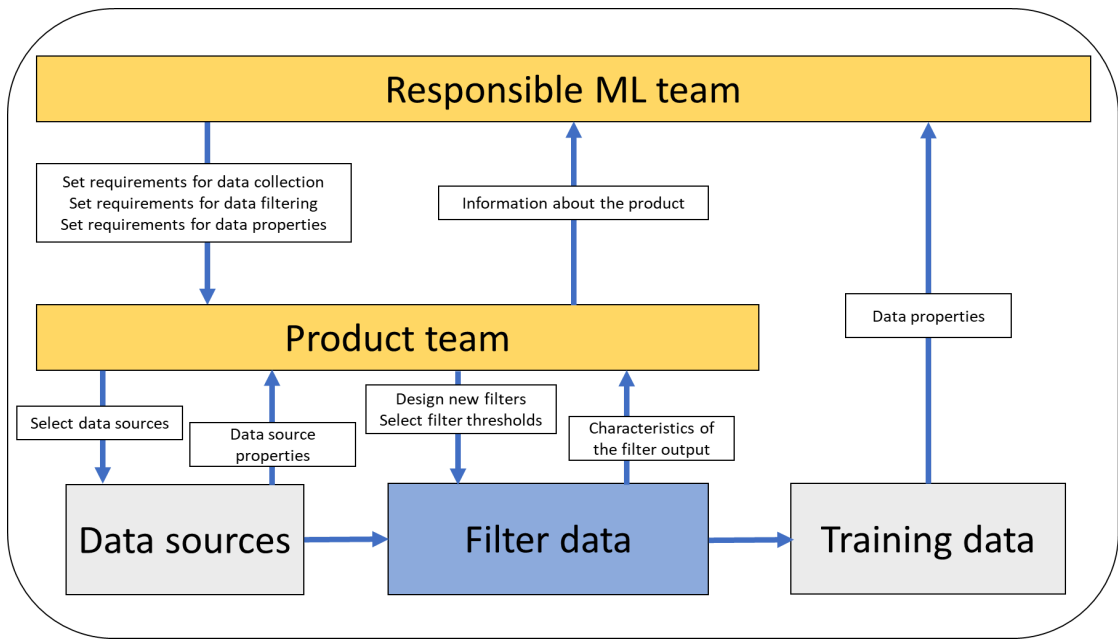
The next STPA step is to identify hazards by considering these losses in relation to the previously identified goal of training data processing. We developed three potential hazards:

- **H1:** System is unable to create a training dataset that contains good quality text-image pairs.
- **H2:** System is unable to create a training dataset that does not contain harmful content.
- **H3:** System is unable to create a training dataset with a diverse representation.

ID	Hazards	Associated Losses					
		L1	L2	L3	L4	L5	L6
H1	System is unable to create a training dataset that contains good quality text-image pairs	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE
H2	System is unable to create a training dataset that does not contain harmful content	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE
H3	System is unable to create a training dataset with a diverse representation	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE

Step 2: Create control structure diagram

Then, we model the control structure. A potential control structure configuration for the data processing stage, as illustrated in Figure 3, includes two organizational/human controllers and one automated controller. We selected the "responsible ML team" and "product team" as human controllers because they can make key decisions in the data processing scope. For this analysis, we assume the responsible ML team is in charge of determining the parameters for a good training dataset and identifying key ethical and legal considerations. The product team is responsible for developing the models that underpin the T2I demo artists will use. We selected as "filter data" an automated controller to reflect where all of the automated filtering operations take place. We identified 6 control actions, denoted in the boxes with down arrows in Figure 3.



For our analysis, we focused on 4 of the 6 control actions occurring between the three controllers and data sources including 1) Set requirements for data filtering, 2) Set requirements for dataset properties, 3) Design new filters and 4) Select filter thresholds. We chose to focus on these 4 control actions because they specifically focus on data processing as opposed to source data collection.

Step 3: Identify unsafe control actions

The STPA analysis for the training data processing stage identified 22 potential Unsafe Control Actions (UCAs) that could lead to three key system hazards: (H1) low-quality text-image pairs, (H2) inclusion of harmful content, and (H3) lack of diverse representation in the training dataset. These UCAs span both human-to-human coordination and human-to-automation interactions, revealing a range of sociotechnical vulnerabilities in the data processing pipeline.

A central pattern in the UCAs involved coordination breakdowns between the responsible ML team and the product team. Multiple UCAs describe cases where the responsible ML team either does not provide, or provides too late, the requirements for data filtering and dataset properties. These lapses were linked to *all three hazards*, highlighting the importance of internal timelines and communication infrastructure for safety and fairness outcomes.

On the automation side, the product team's design and parameterization of data filters—particularly for non-English content, adult content, and other sensitive categories—were also common sources of unsafe actions. Overly aggressive filters risked excluding a disproportionate amount of data, leading to dataset sparsity and reduced representational diversity. Conversely, under-performing filters failed to catch inappropriate or harmful material, raising risks to both system performance and user safety.

Importantly, several UCAs also revealed underlying structural assumptions in technical design, such as an over-reliance on English-speaking content or narrow definitions of harmful material. These UCAs demonstrate how implicit values embedded in filtering logic can shape the resulting training data—and by extension, the outputs of the T2I system—in ways that undermine creativity, accessibility, and equity.

Collectively, the 22 UCAs show how ethical and functional risks emerge not just from model outputs, but from upstream decisions and omissions during data preparation. The STPA framework enabled the identification of these risks at the organizational and procedural levels, offering a practical method for embedding safety and responsibility into ML data pipelines from the outset.

Step 4: Identify loss scenarios

The STPA analysis surfaced a set of plausible loss scenarios that reveal how unsafe control actions—especially those involving coordination breakdowns between human actors—can lead to hazards across quality, safety, and equity dimensions in T2I training data.

A recurring theme across these scenarios is misalignment in communication and timelines between the responsible ML team and the product development team. For instance, when the product team fails to clearly articulate the intended use case and end-user context, the responsible ML team must make assumptions or use generic guidelines. This can result in filters or dataset requirements that are misaligned with the actual context of use—potentially leading to loss of diversity (L2) or loss of accessibility (L3) if, for example, culturally specific or marginalized representations are omitted due to inappropriate filtering criteria.

Another class of scenarios involves delayed or incomplete feedback loops. In several cases, the responsible ML team is overloaded with concurrent review requests and cannot deliver requirements on time. As a result, the product team may move forward with data processing using default or outdated filters, leading to datasets that include harmful content (H2) or lack diverse representation (H3). In reverse, if the product team withholds contextual information until late in the process, the responsible ML team cannot meaningfully tailor safeguards.

These loss scenarios also highlight missing or inadequate policies for identifying appropriate requirements. In the absence of established guidelines, the responsible ML team may resort to ad hoc solutions or default to internal brainstorming—introducing inconsistency and reducing rigor in content filtering. This contributes to losses of quality (L5) and reputation (L6) if the resulting dataset enables problematic model outputs.

Finally, many scenarios emerge from an overreliance on assumed defaults—e.g., assuming that product timelines always take precedence, or that ML teams can generalize across use cases. These assumptions break down in high-stakes or sensitive applications, revealing the need for explicit timing coordination, shared mental models of responsibilities, and sustained investment in process-level safety measures.

Together, these loss scenarios underscore how sociotechnical failures—such as poor internal communication or unclear governance—can propagate through seemingly mundane development processes, with significant downstream consequences for T2I model safety and trustworthiness

Scope 2 - model integration

Many AI ethics assessments and audits focus on one ML model [25, 76]. However, in production, multiple models often are used to achieve the intended functions for a given product or demo [83]. We conducted STPA to examine the integration of a T2I model in a productionized demo, such as those released by Stability AI [2] and OpenAI [63]. From our interviews with T2I developers and existing literature [54, 64], we identified demos often include at least three types of models: (1) an input prompt classifier (which either block or filter the text prompt), (2) the T2I model, and (3) output image classifiers (which either block or filter the generated image). For any given T2I demo, there could be multiple classifiers employed for filtering text prompts and images. For the purposes of this illustrative analysis, we assume there are only two classifiers. One for "adult" content run on the input text prompt and one for analyzing the generated image.

Step 1: Identify purpose of analysis

Losses:

From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.**

Stakeholders: From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68]

Values: Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist's reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company's reputation.

ID Loss

- L1 Loss of creativity
- L2 Loss of diversity in generated image
- L3 Loss of accessibility (who can use these systems)
- L4 Loss of efficiency (in generating image)
- L5 Loss of quality
- L6 Loss of reputation

System boundary: Here the goal of the system is to create a T2I demo for public and creative use.

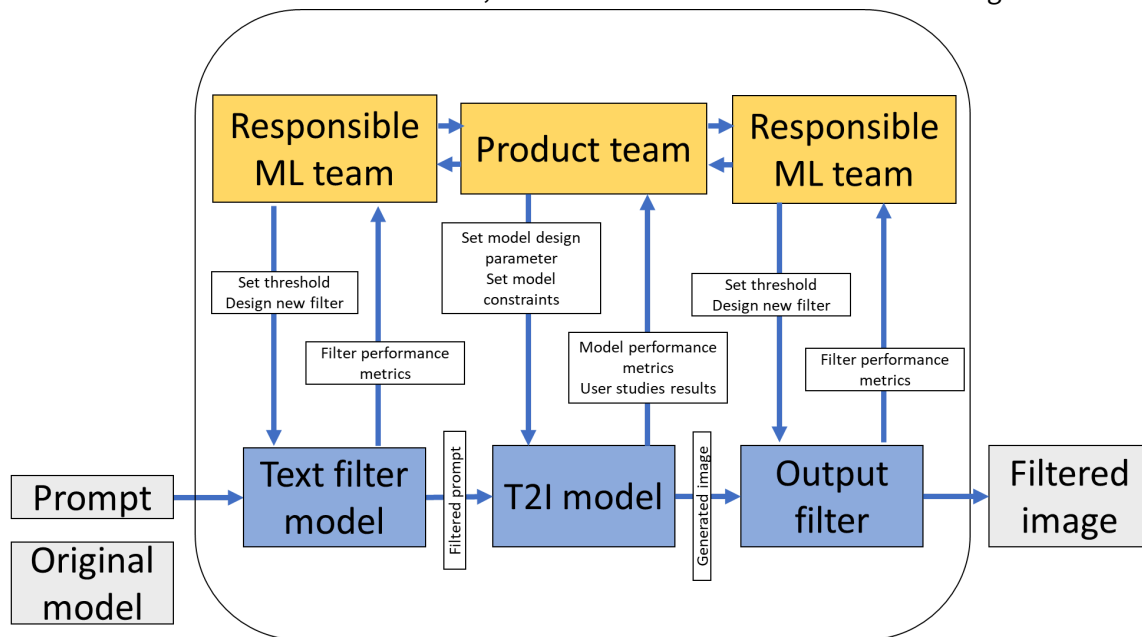
System-level hazards:

- H1: System creates an image that does not match the prompt.
- H2: System cannot generate an image.
- H3: System generates an unsafe image (i.e., with adult content).

ID	Hazards	Associated Losses					
		L1	L2	L3	L4	L5	L6
H1	System creates an image that does not match the prompt.	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE
H2	System cannot generate an image.	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE
H3	System generates an unsafe image	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE

Step 2: Create control structure diagram

A potential configuration of the control structure (as illustrated in Figure 4) for this scope of analysis includes two organizational/ human controllers and three automated controllers. The organizational/human controllers are the "responsible ML team" and the "product team." The automated controllers are the input text classifier, the T2I model, and the output image classifier. We identified 6 control actions, denoted in boxes with down arrows in Figure 4.



Step 3: Identify unsafe control actions

The STPA for Scope 2 identified 16 Unsafe Control Actions (UCAs) related to the integration of multiple machine learning models within a T2I demo. These UCAs spanned both organizational processes and technical configurations across a system composed of at least three models: an input prompt classifier, the T2I model itself, and an output image classifier.

Many UCAs emerged from coordination breakdowns between the responsible ML team and the product team, such as delays in setting model constraints, unclear or incomplete filtering requirements, and misaligned timelines. For example, a UCA where the product team fails to implement appropriate filter thresholds for text prompts can result in the model rejecting legitimate prompts, thereby contributing to H2: system cannot generate an image or H3: generation of unsafe content.

Other UCAs centered around incorrect or poorly calibrated model parameters. These include scenarios where inappropriate constraints are placed on the T2I model version used in the demo, or where the responsible ML team designs overly restrictive filters that limit creative expression. These actions often had dual consequences: they reduced the system's ability to fulfill its intended creative purpose (H1: mismatch between prompt and image) and introduced ethical or reputational risks (L1: loss of creativity, L6: loss of reputation).

Critically, several UCAs revealed missing feedback loops—for instance, the product team does not receive feedback from the output image classifier, leading to a misalignment between model behavior and system goals. These gaps highlight structural limitations in how model integration decisions are made and updated across teams, particularly when responsibilities for safety and performance are fragmented.

The 16 UCAs illustrate how integrating multiple models into a cohesive product introduces unique hazards not visible when evaluating any one model in isolation. These include emergent risks at the intersection of classifier behavior, model constraints, and organizational workflow—risks that conventional single-model assessments or static checklists may fail to capture.

Step 4: Identify loss scenarios

The loss scenarios identified in Scope 2 demonstrate how failures in coordination, timing, and contextual understanding across organizational teams and model components can result in unsafe, inaccessible, or low-quality T2I outputs. These scenarios reflect not only technical missteps but also broader sociotechnical breakdowns in how safety, creativity, and usability are managed in product development.

A recurring pattern in the scenarios involves reuse or misconfiguration of pre-existing filter thresholds. For example, when the responsible ML team (RMT) fails to update thresholds for input or output filters (Scenarios 1 and 7), old parameters—possibly tuned for different use cases—are applied inappropriately. This leads to mismatches between the system's behavior and the actual user context, contributing to hazards such as the generation of irrelevant or blocked images (H1, H2) and losses like L1: creativity and L6: reputation. These failures stem from limited time, unclear ownership of filter configurations, or lack of technical knowledge for modifying embedded components.

Another critical set of scenarios centers on inappropriate filtering logic—either due to poor validation (Scenario 2) or misalignment with user intent (Scenario 4). In these cases, filters either fail to capture problematic content or overly restrict valid, expressive inputs. For instance, filters might exclude prompts common in marginalized communities or artistic experimentation, leading to L2: loss of diversity and L3: loss of accessibility.

Several scenarios illustrate how timing failures between teams undermine safety and effectiveness. For instance, when RMT delivers new prompt or image filters too late (Scenarios 3 and 8), the product team must launch with incomplete or outdated safeguards. These timing mismatches are often due to cycle time constraints, unclear prioritization, or overloaded review queues—problems deeply embedded in organizational workflows.

Finally, loss scenarios associated with model parameter setting and constraint definition reveal how cautious over-restriction by the product team (e.g., limiting colors or shapes, as in Scenario 6) may constrain user creativity, leading to L1, L4, and L5 losses. These decisions are often made out of legal or reputational concerns but may inadvertently alienate user groups or limit the system's artistic utility.

Together, these loss scenarios show that integration-stage hazards emerge from more than just technical flaws—they stem

from misaligned expectations, insufficient feedback loops, and embedded organizational trade-offs. These failures compromise not only functional performance but also the system's capacity to serve diverse users in creative and socially responsible ways.

Scope 3 - use

This scope of analysis focuses on how an artist would use a T2I demo as part of their creative practice. The information we gathered from the artist workshops heavily informs this scope. To perform this illustrative STPA analyses, we assume the artist is a filmmaker, and they are creating a video for a client. The key stakeholders include the client, the filmmaker, and the public/viewer of the video. The scope focuses on how the filmmaker uses the T2I demo as a tool for generating images in their storyboard mock-up, where they visualize and share ideas with the client

Step 1: Identify purpose of analysis

Losses:

From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.**

Stakeholders: From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68]

Values: Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist's reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company's reputation.

ID Loss

- | | |
|----|---|
| L1 | Loss of creativity |
| L2 | Loss of diversity in generated image |
| L3 | Loss of accessibility (who can use these systems) |
| L4 | Loss of efficiency (in generating image) |
| L5 | Loss of quality |
| L6 | Loss of reputation |

System boundary: The goal of our system is to support an artist in creating a video for a client using a T2I demo.

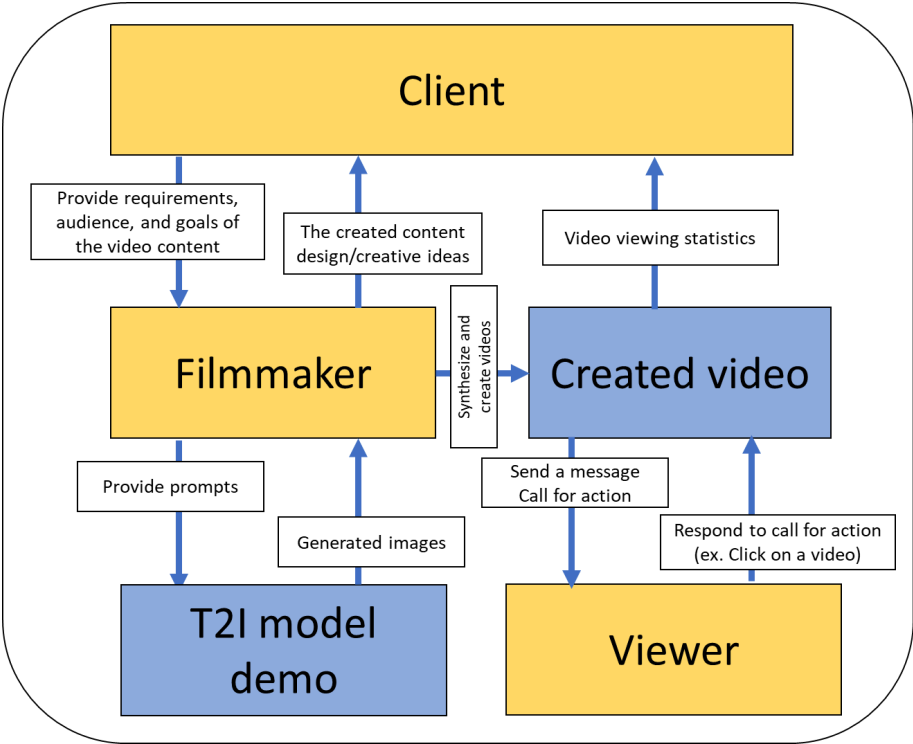
System-level hazards:

- H1: The artist cannot create a video. (L1, L3, L6)
- H2: The artist cannot create a video that meets client requirements. (L1, L3, L5, L6)
- H3: The video disseminates false and harmful information. (L2, L3, L5, L6)

ID	Hazards	Associated Losses					
		L1	L2	L3	L4	L5	L6
H1	The artist cannot create a video	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE
H2	The artist cannot create a video that meets client requirements.	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE
H3	The video disseminates false and harmful information.	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE

Step 2: Create control structure diagram

A potential configuration of the control structure (illustrated in Figure 5 includes three human controllers (i.e., the client, filmmaker, and the viewer) and two automated controllers (i.e., the T2I model demo and the created video). For the purpose of this analysis, we focus on the 3 control actions between the 3 human controllers and the T2I model demo (denoted in boxes with down arrows in Figure 5): (1) Provide requirements/goals/audience for the video content; (2) provide prompts; (3) send a message.



Step 3: Identify unsafe control actions

The analysis identified 9 Unsafe Control Actions across interactions between the client, filmmaker, viewer, and T2I system. These UCAs reflect both communication failures and limitations in system behavior that constrain creative expression or introduce harm.

Several UCAs involved miscommunication or misalignment of expectations between the client and filmmaker. These include vague or missing project goals, which result in a video that misrepresents the client's intent or fails to meet their standards (linked to H2). Others reflect the challenges artists face in generating effective prompts for T2I systems—either due to a lack of domain alignment (e.g., the system can't interpret the prompt) or model bias, which disproportionately affects artists

from marginalized backgrounds (H1, H2, H3).

Additional UCAs occurred during the viewer's interaction with the final product. Inaccurate or harmful interpretations of the video (H3) can emerge due to ambiguous visuals or misinformation generated by the T2I model. Viewers may also fail to respond or engage if the message is unclear or unrelatable (H1, H3).

Overall, the UCAs point to critical sociotechnical gaps—from prompt literacy and accessibility to interpretability and equity—that can undermine the effectiveness and safety of T2I systems in real-world creative workflows.

Step 4: Identify loss scenarios

Each UCA was linked to 1–2 plausible loss scenarios that illustrate how control failures translate into real-world consequences. These scenarios highlight the cascading nature of misalignment, bias, and ambiguity in human-AI collaboration.

Some scenarios show how unclear client communication leads the filmmaker to produce a misaligned storyboard, eroding trust and causing reputational or financial loss (L5, L6). Others emphasize how model limitations or prompt misalignment prevent the filmmaker from generating relevant images, resulting in lower-quality outputs and creative inefficiencies (L1, L4).

Importantly, several scenarios spotlight equity issues—such as when artists from marginalized backgrounds receive worse output quality due to biased model training. These harms are not merely technical failures but represent deeper structural issues of accessibility, diversity, and inclusion in ML model development (L2, L3).

Finally, viewer-side scenarios reveal that lack of transparency or clarity in the AI-generated visuals can lead to misinterpretation or misinformation spreading, causing ethical and reputational risks for both the artist and the client (L6).